

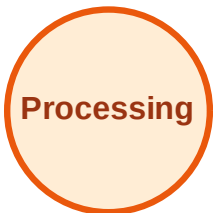
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



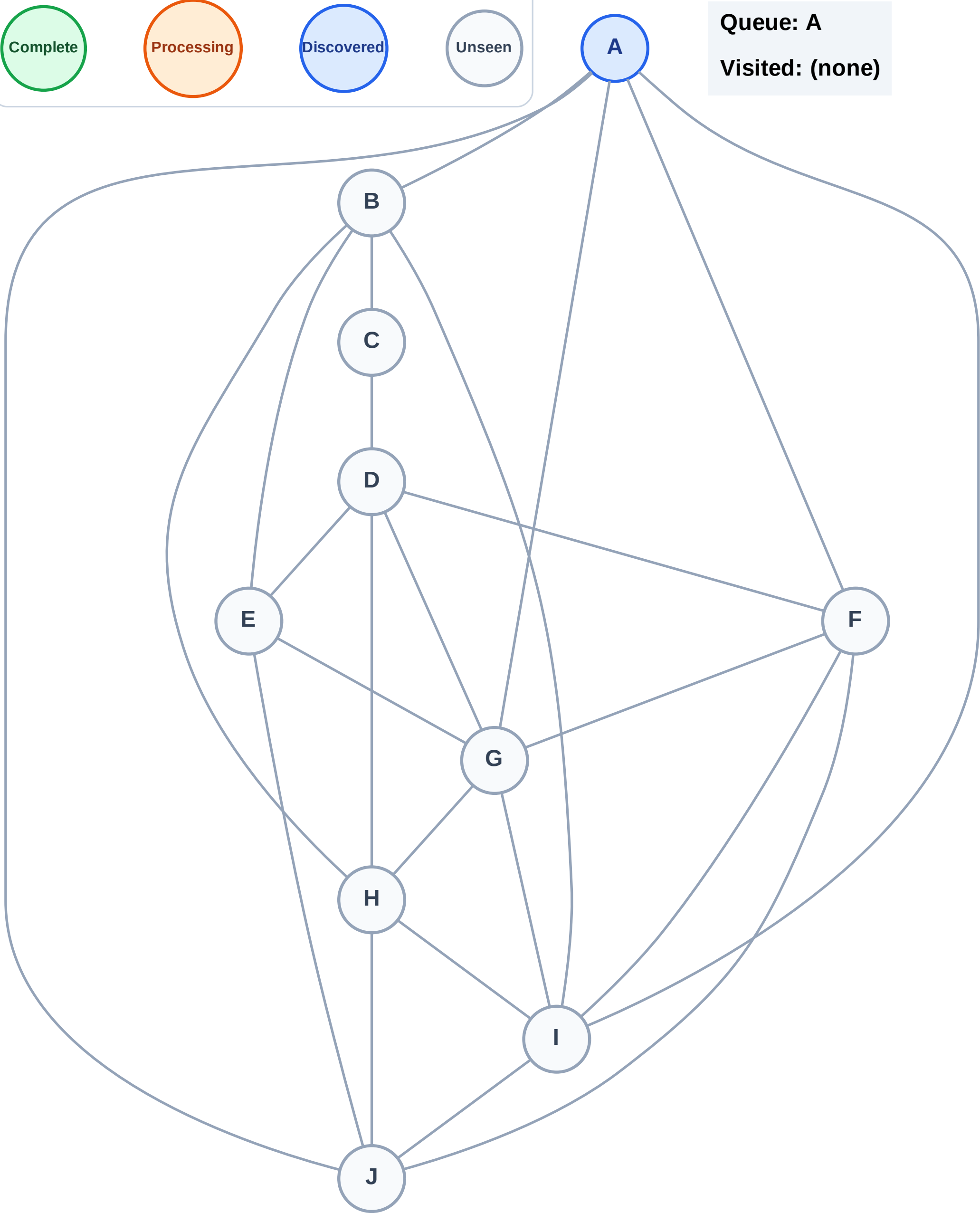
Discovered



Unseen

Queue: A

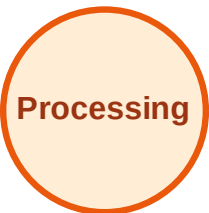
Visited: (none)



BFS Traversal

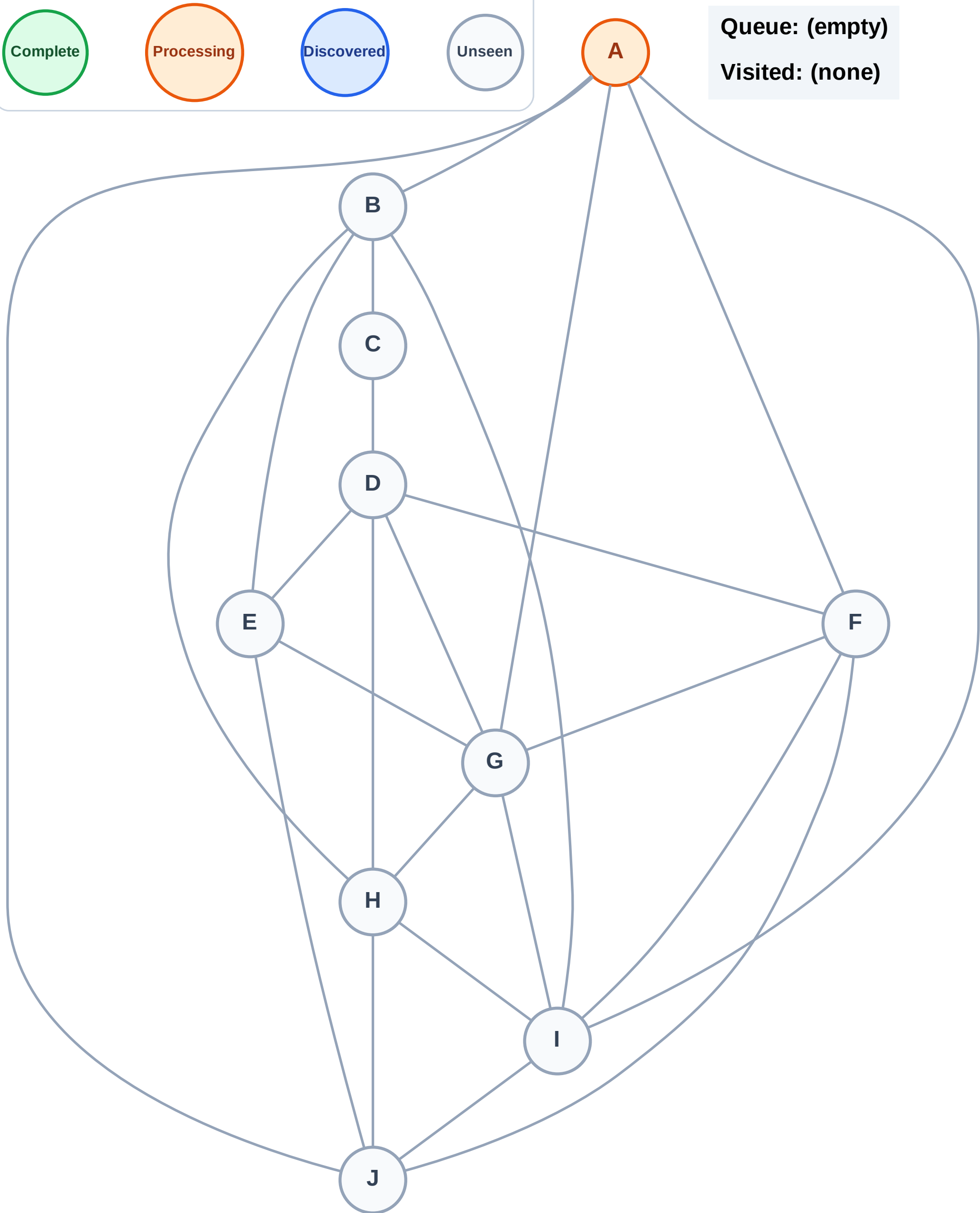
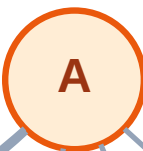
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, F, G, I, J from A

Legend

Complete

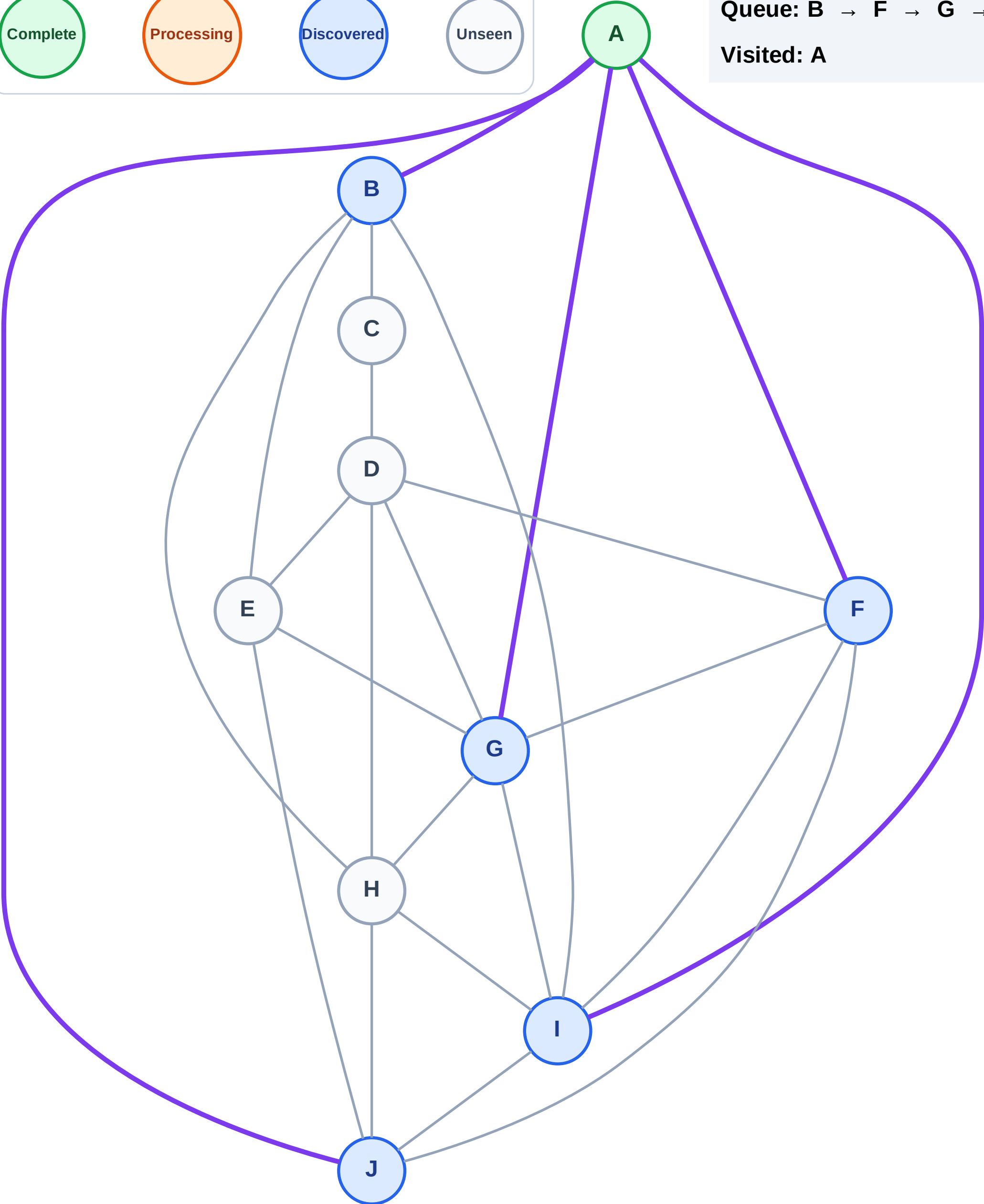
Processing

Discovered

Unseen

Queue: B → F → G → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

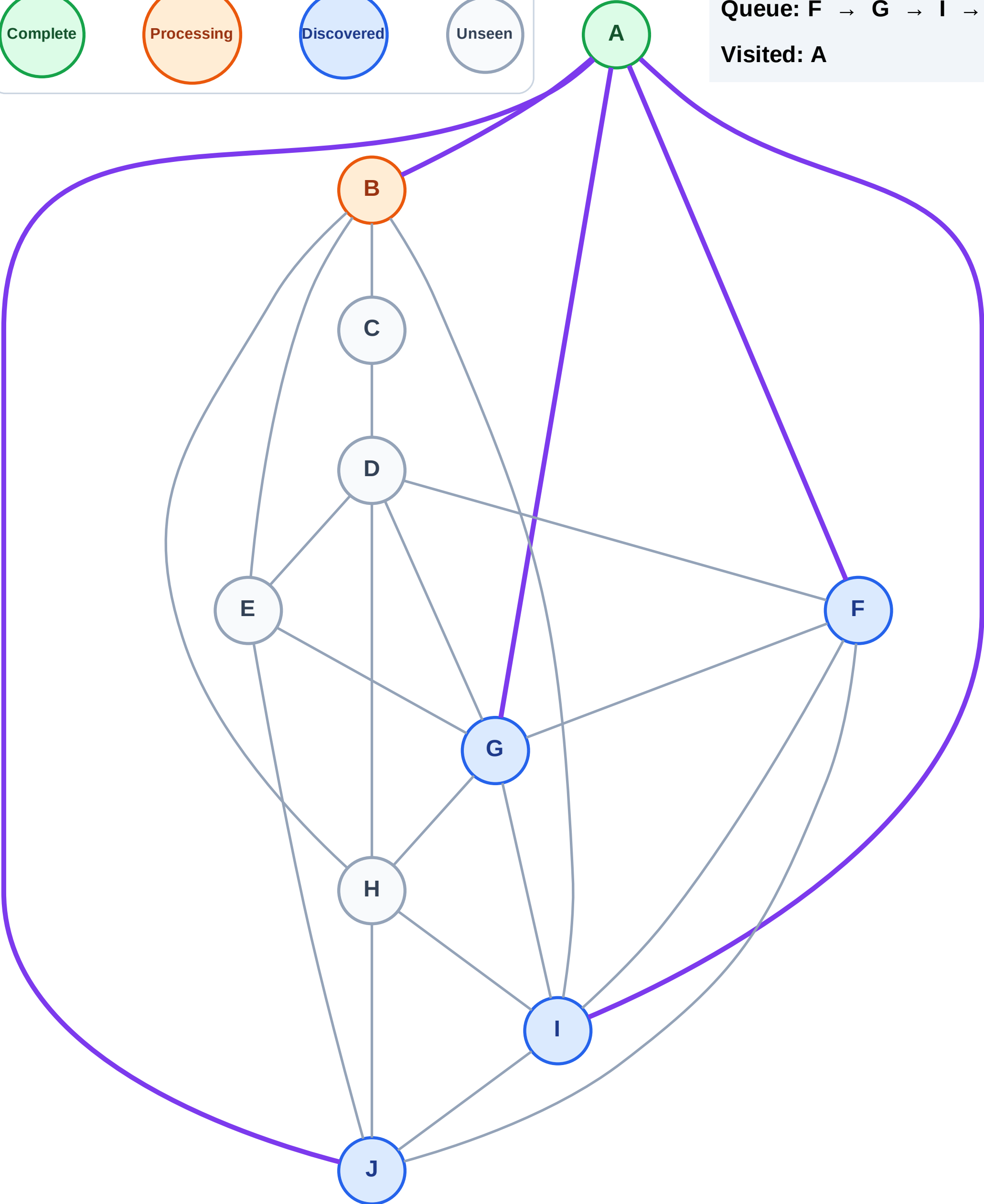
Processing

Discovered

Unseen

Queue: F → G → I → J

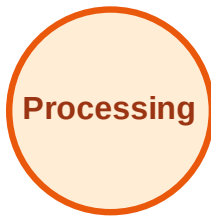
Visited: A



BFS Traversal

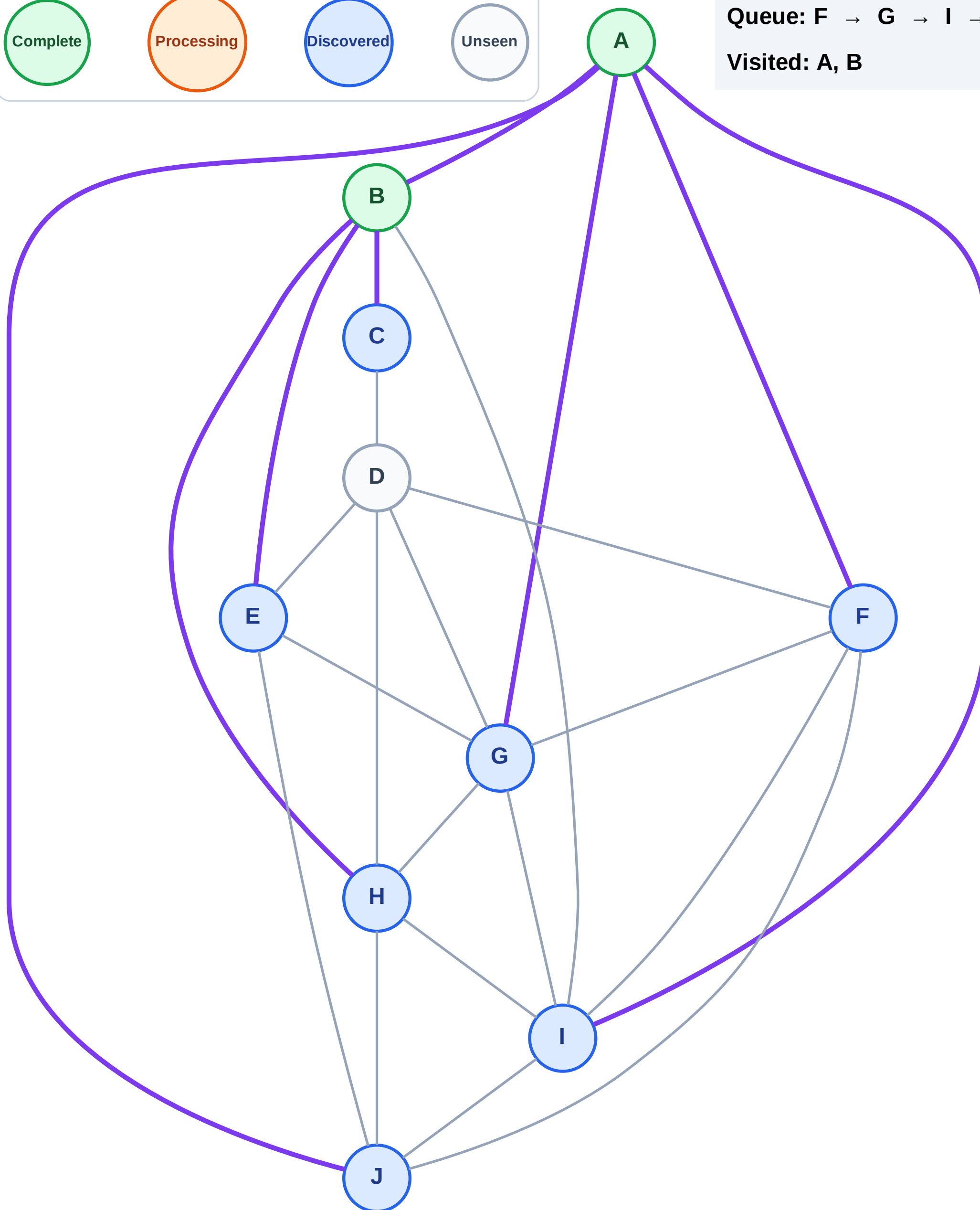
Step 5: Discover C, E, H from B

Legend



Queue: F → G → I → J → C → E → H

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

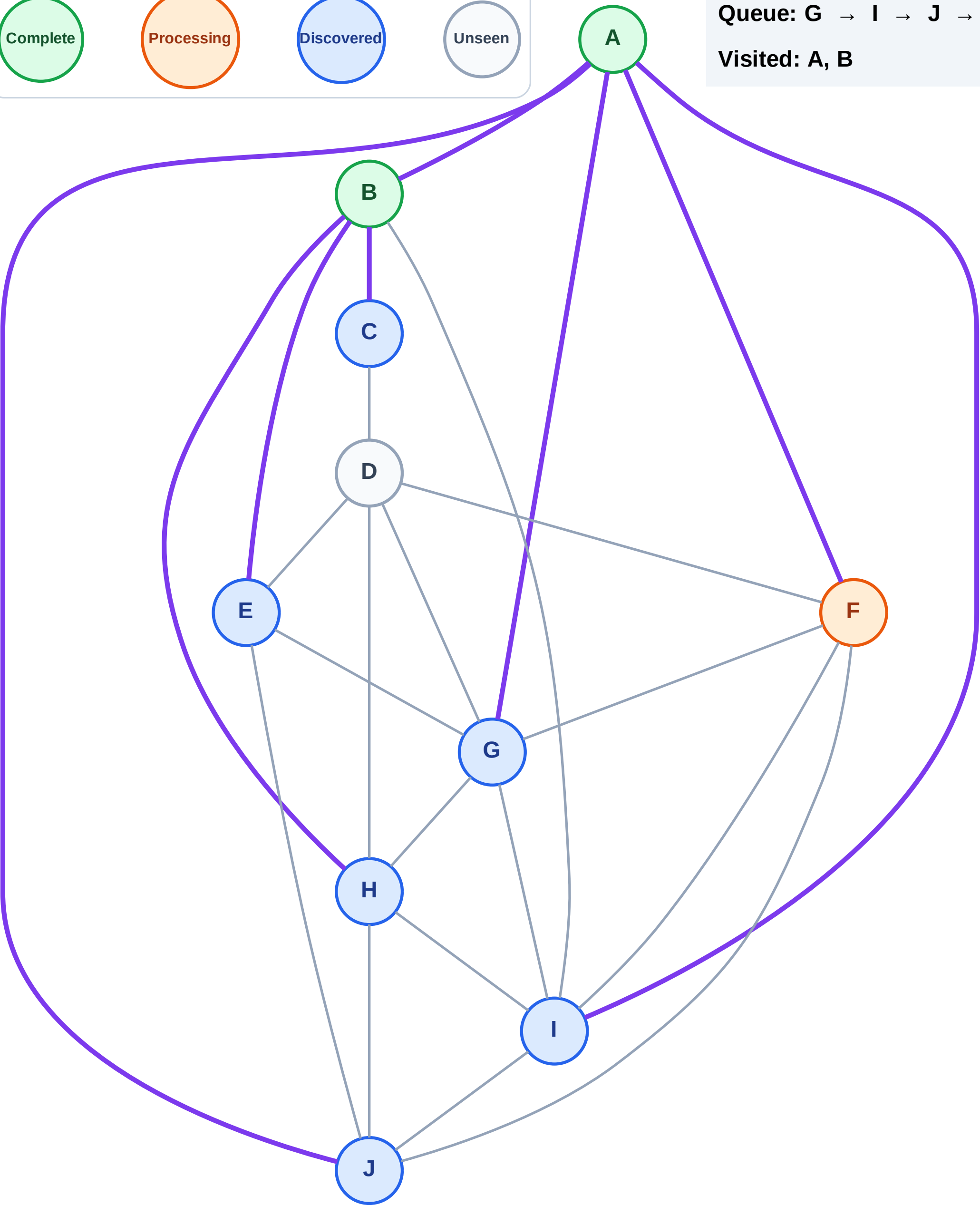
Processing

Discovered

Unseen

Queue: G → I → J → C → E → H

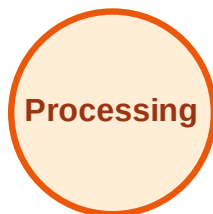
Visited: A, B



BFS Traversal

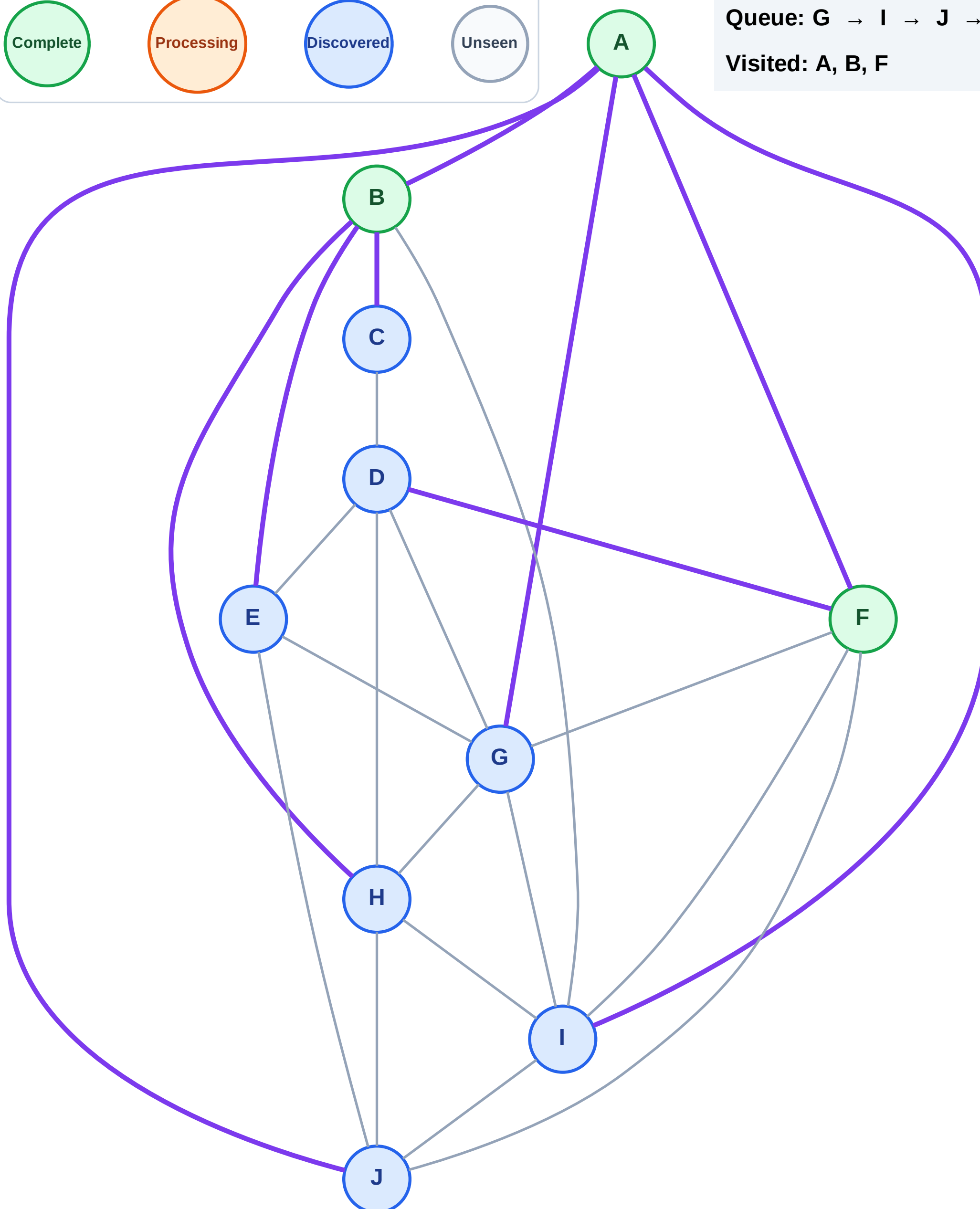
Step 7: Discover D from F

Legend



Queue: G → I → J → C → E → H → D

Visited: A, B, F

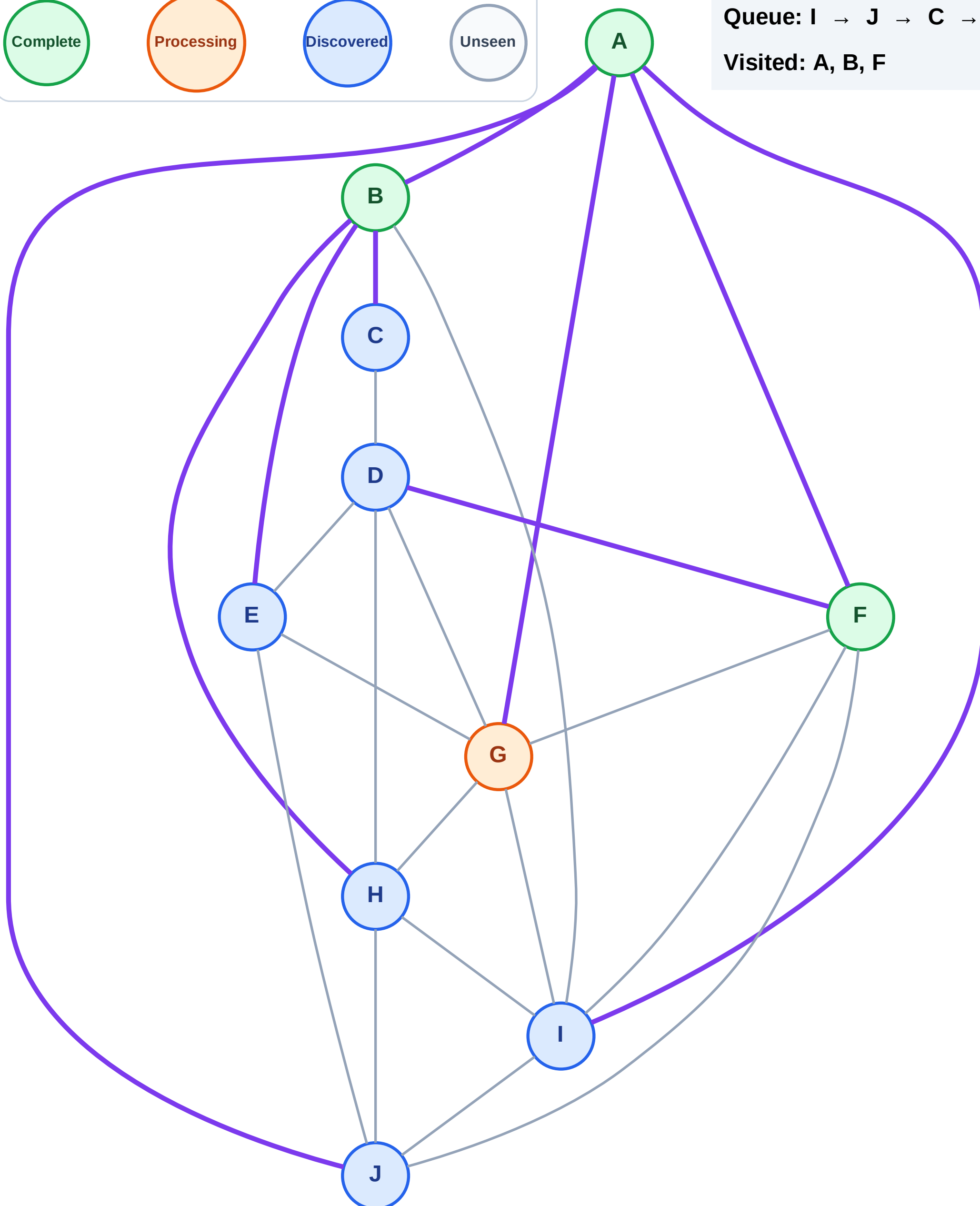


Step 8: Process node G

Legend

Unseen

Visited: A, B, F



BFS Traversal

Step 9: No new neighbors from G

Legend

Complete

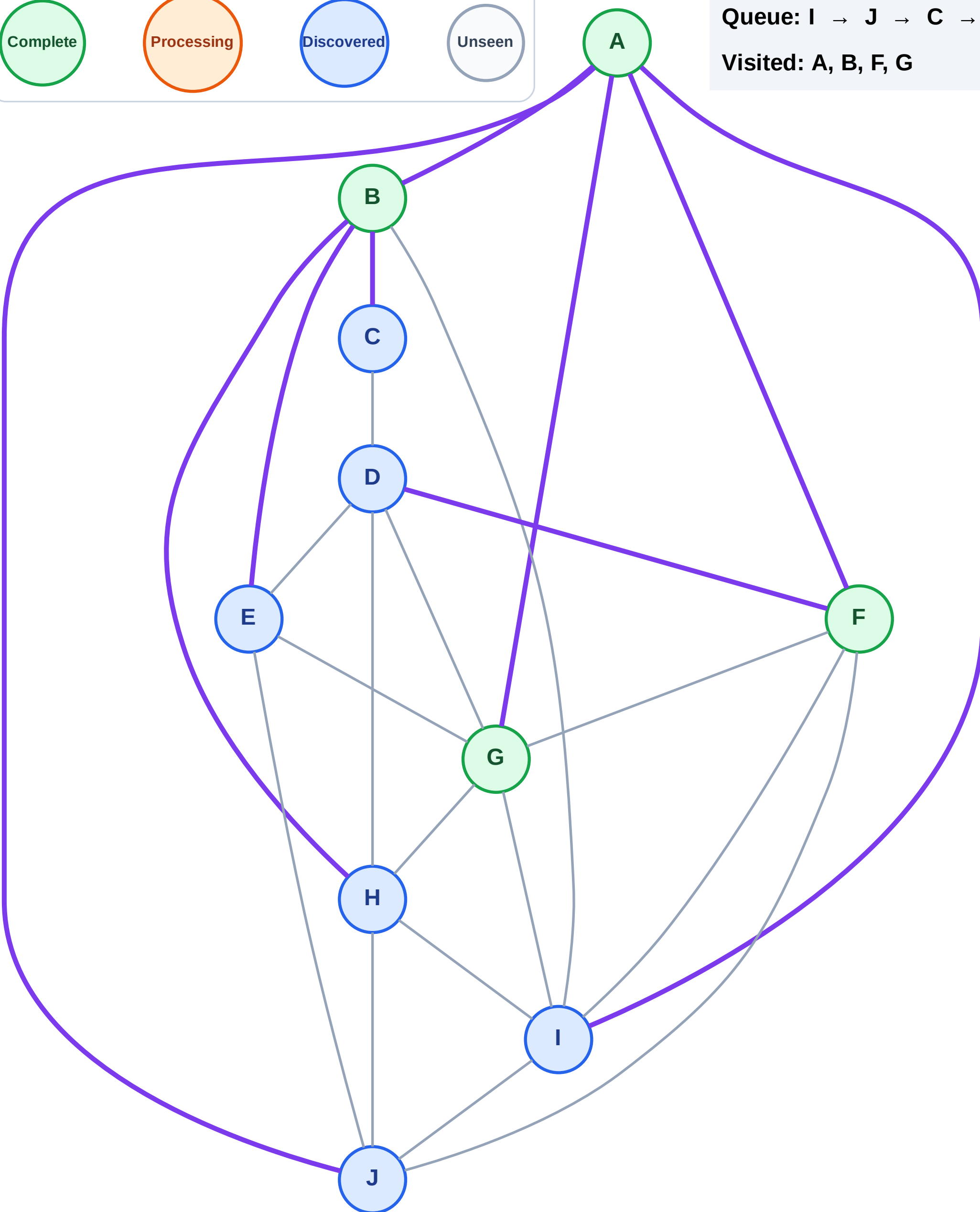
Processing

Discovered

Unseen

Queue: I → J → C → E → H → D

Visited: A, B, F, G



BFS Traversal

Step 10: Process node I

Legend

Complete

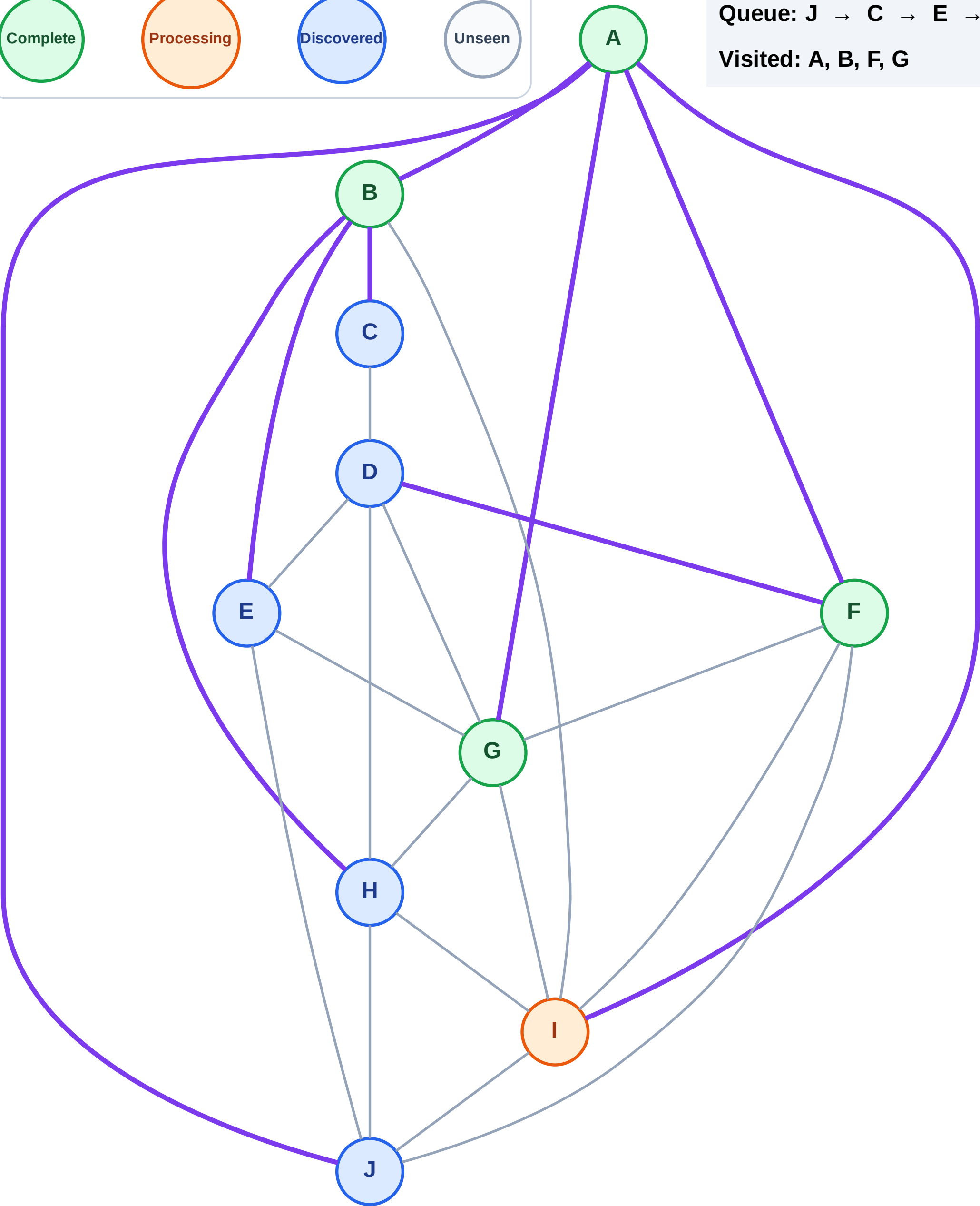
Processing

Discovered

Unseen

Queue: J → C → E → H → D

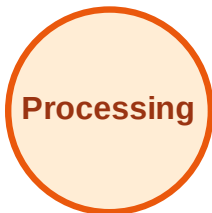
Visited: A, B, F, G



BFS Traversal

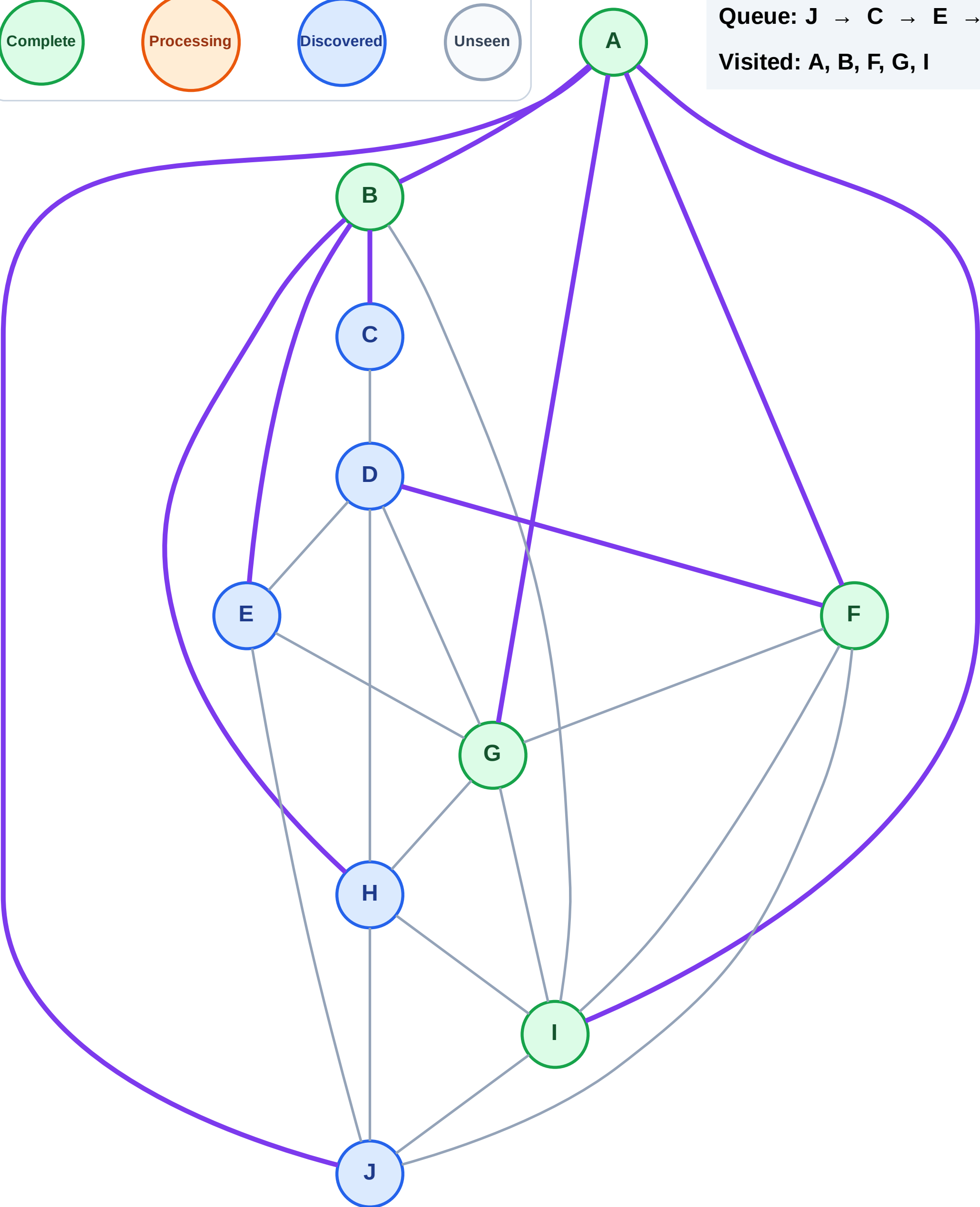
Step 11: No new neighbors from I

Legend



Queue: J → C → E → H → D

Visited: A, B, F, G, I



BFS Traversal

Step 12: Process node J

Legend

Complete

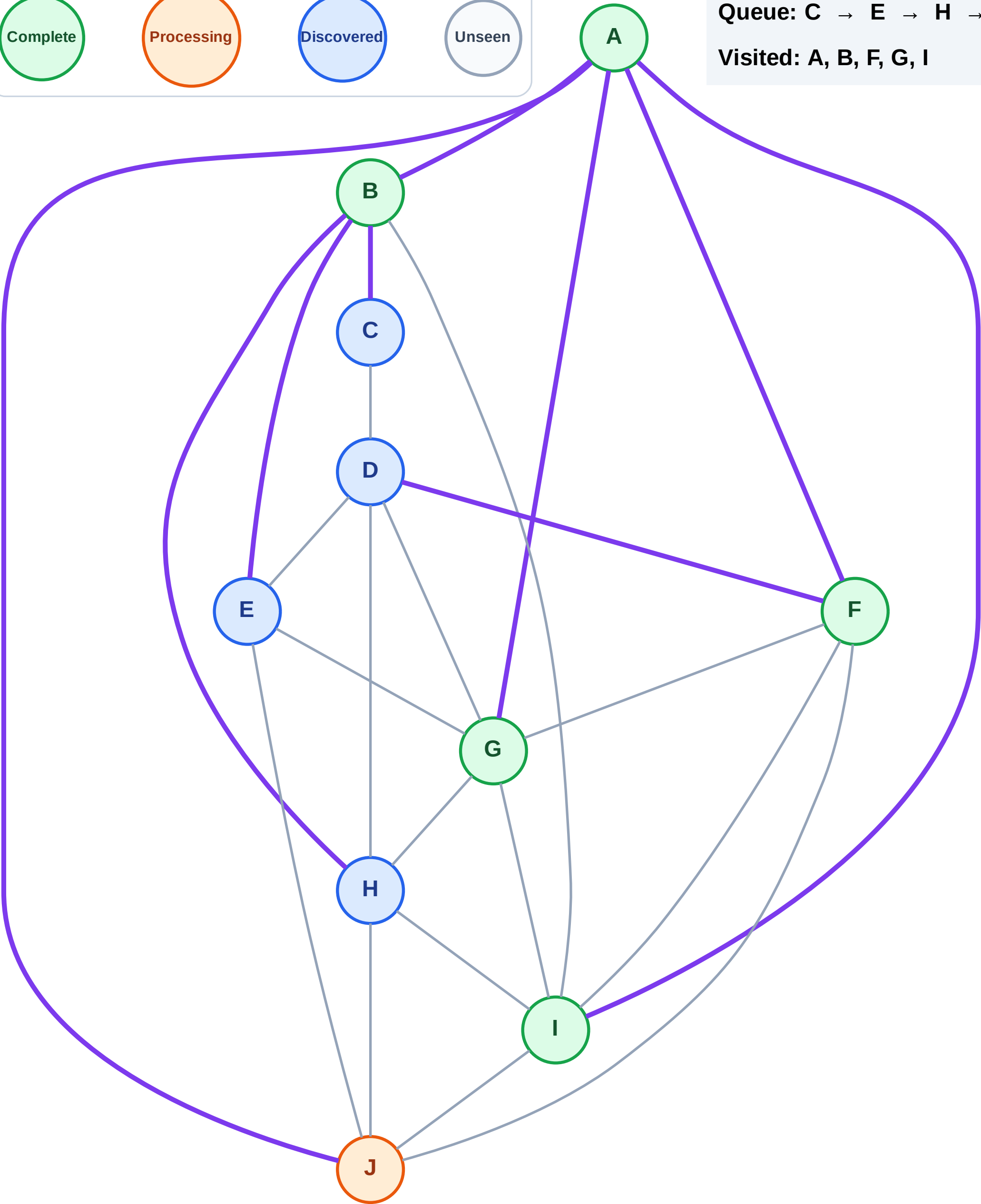
Processing

Discovered

Unseen

Queue: C → E → H → D

Visited: A, B, F, G, I



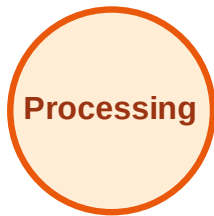
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



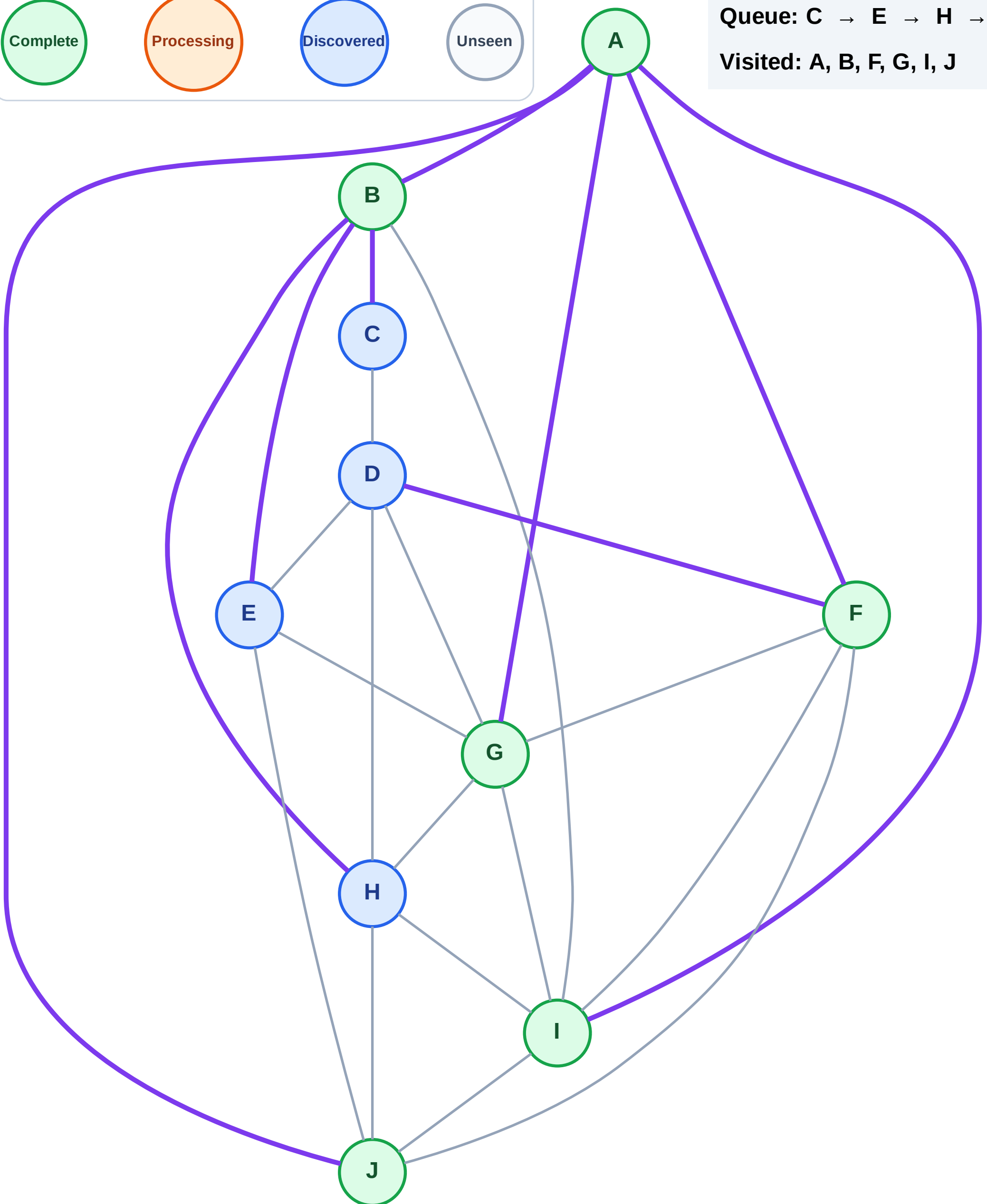
Discovered



Unseen

Queue: C → E → H → D

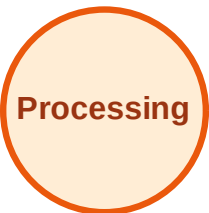
Visited: A, B, F, G, I, J



BFS Traversal

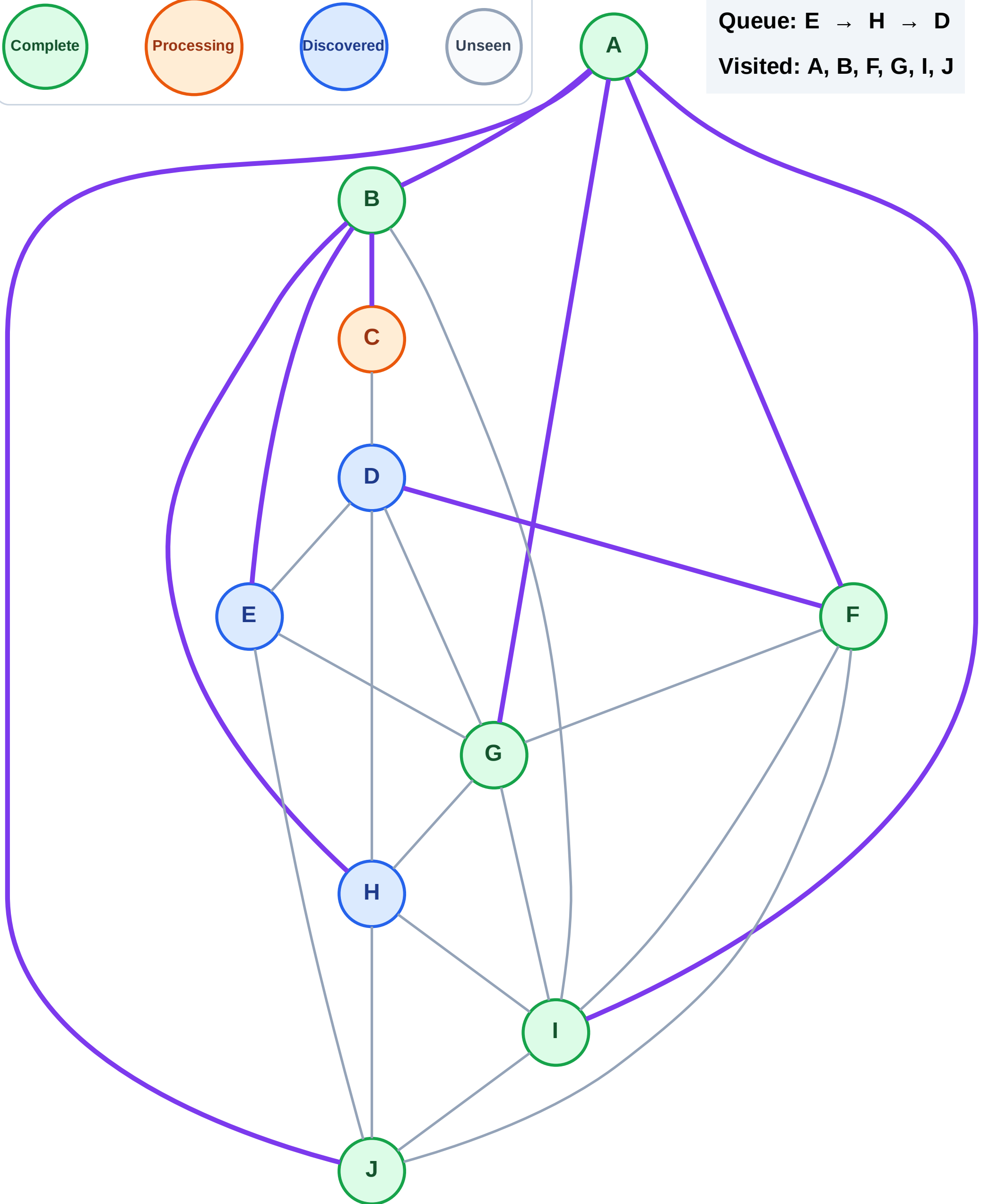
Step 14: Process node C

Legend



Queue: E → H → D

Visited: A, B, F, G, I, J



BFS Traversal

Step 15: No new neighbors from C

Legend

Complete

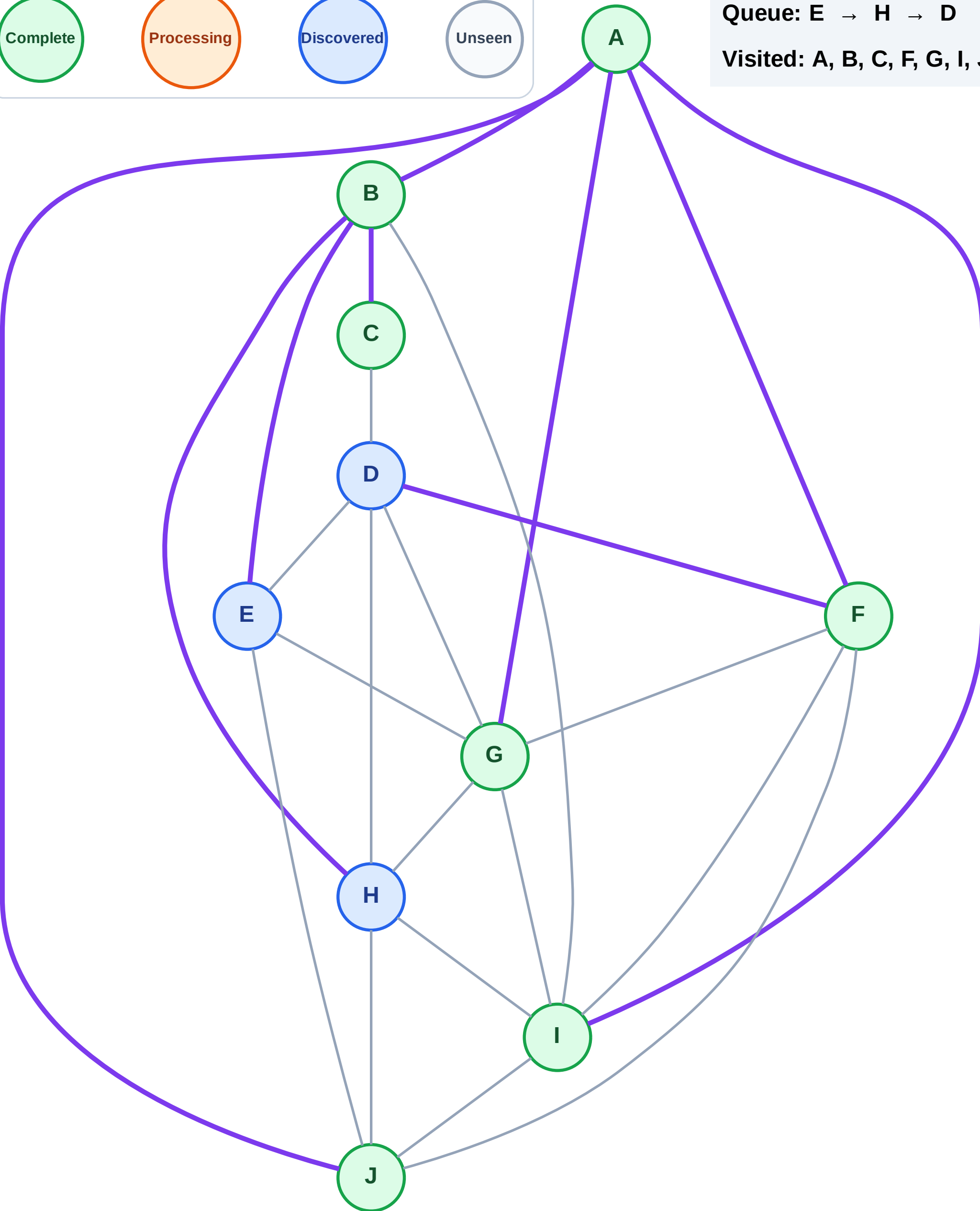
Processing

Discovered

Unseen

Queue: E → H → D

Visited: A, B, C, F, G, I, J



BFS Traversal

Step 16: Process node E

Legend

Complete

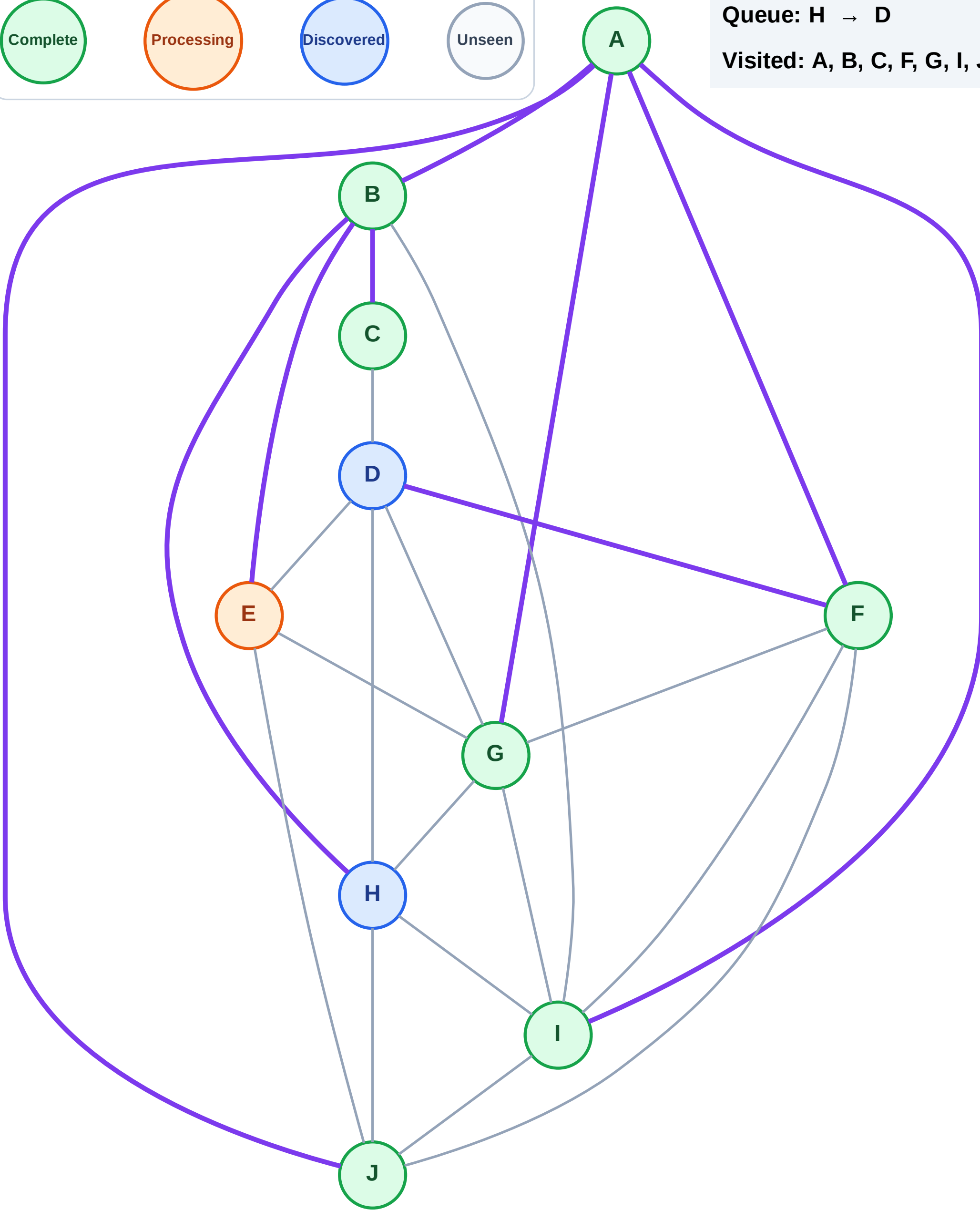
Processing

Discovered

Unseen

Queue: H → D

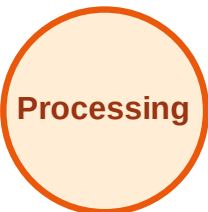
Visited: A, B, C, F, G, I, J



BFS Traversal

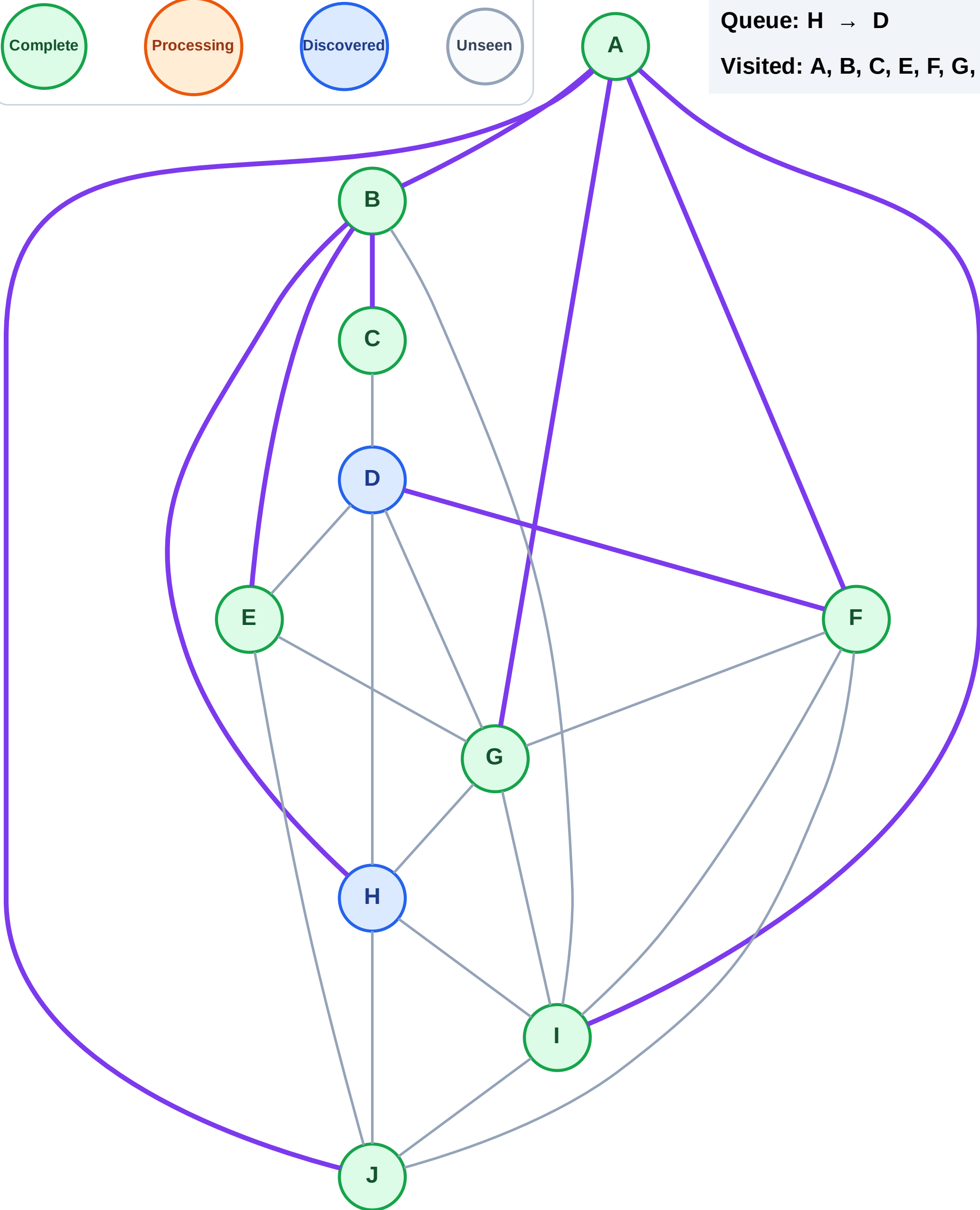
Step 17: No new neighbors from E

Legend



Queue: H → D

Visited: A, B, C, E, F, G, I, J



BFS Traversal

Step 18: Process node H

Legend

Complete

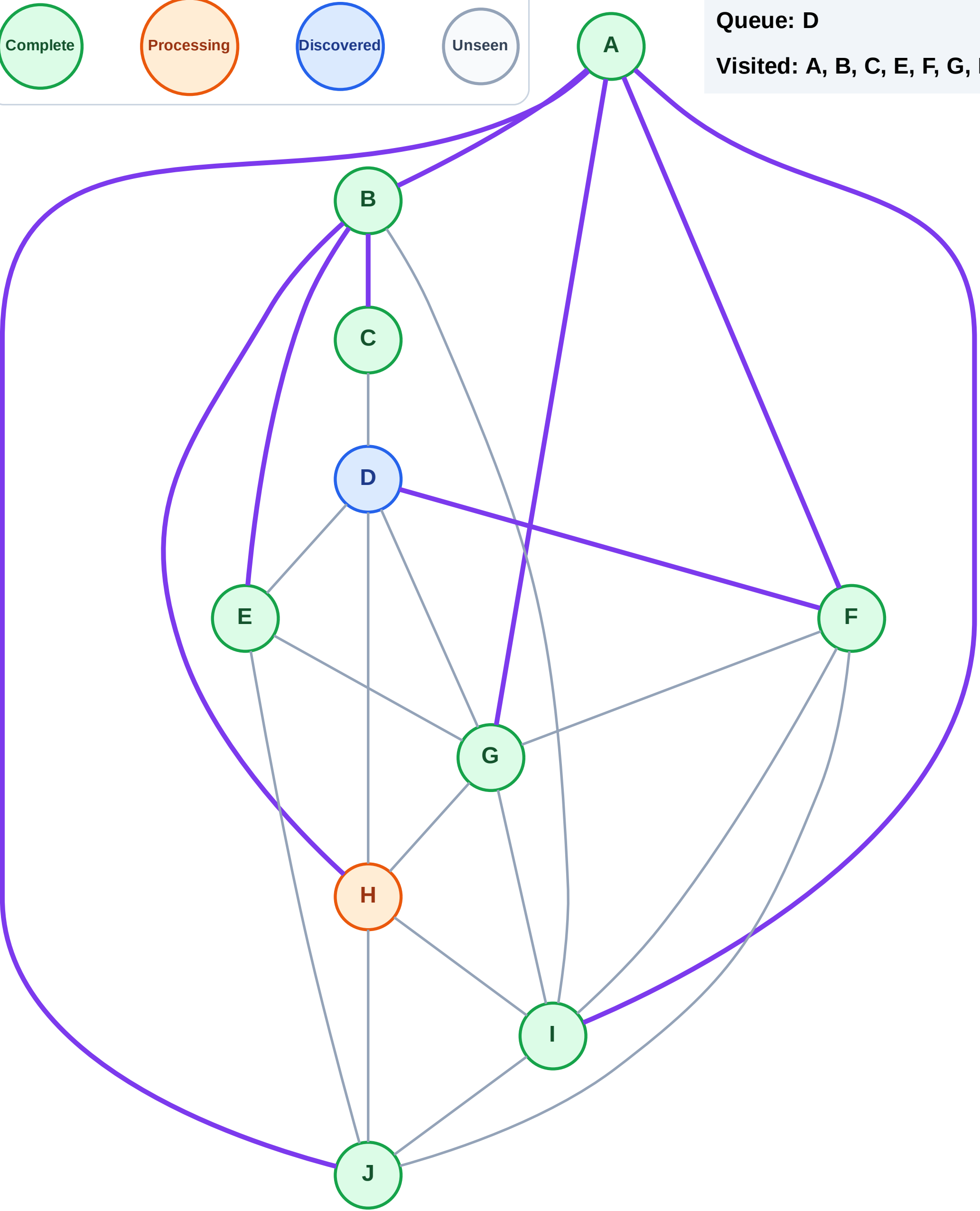
Processing

Discovered

Unseen

Queue: D

Visited: A, B, C, E, F, G, I, J



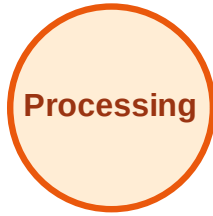
BFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



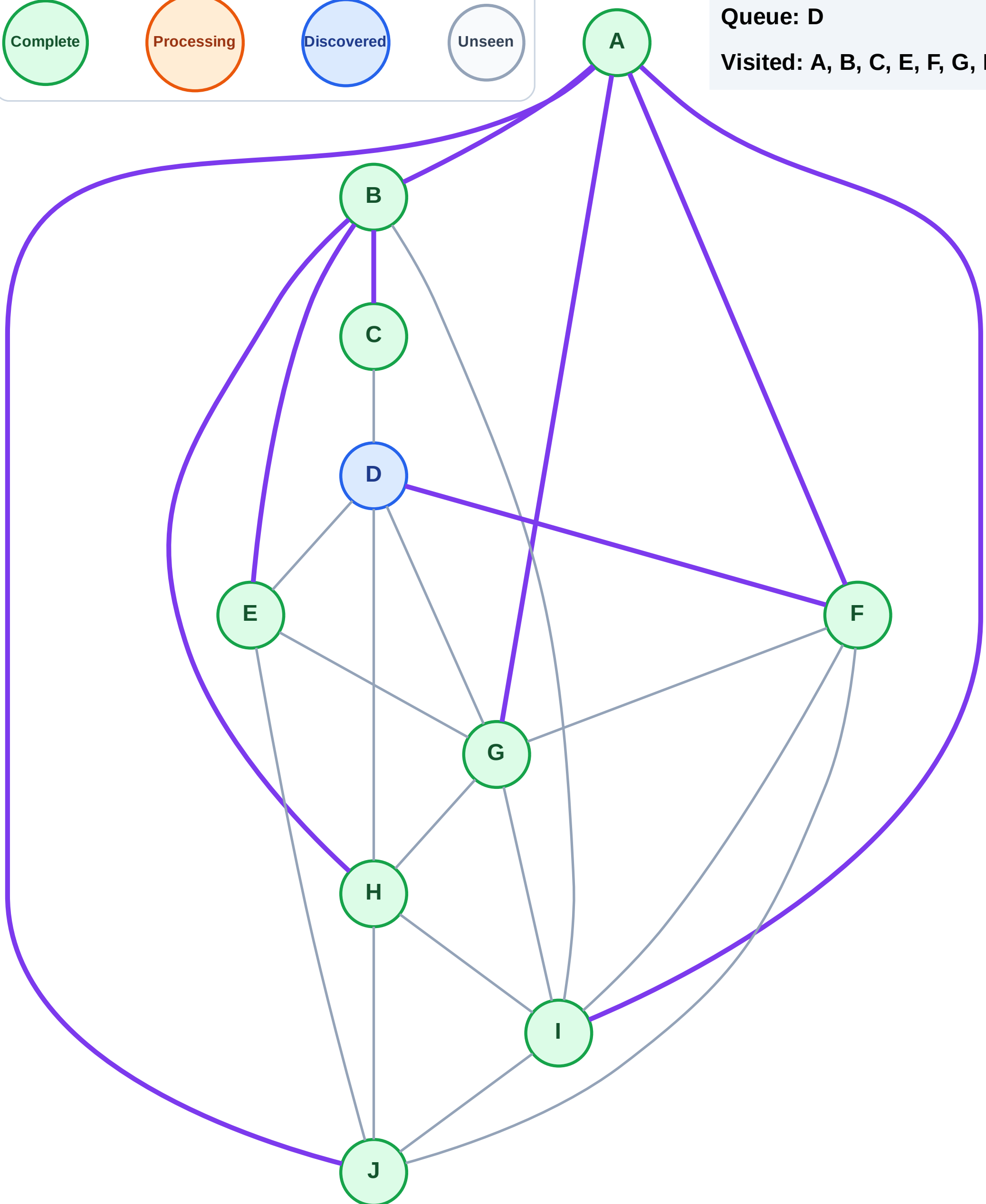
Discovered



Unseen

Queue: D

Visited: A, B, C, E, F, G, H, I, J



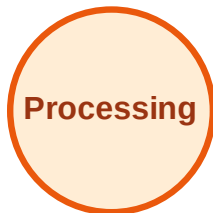
BFS Traversal

Step 20: Process node D

Legend



Complete



Processing



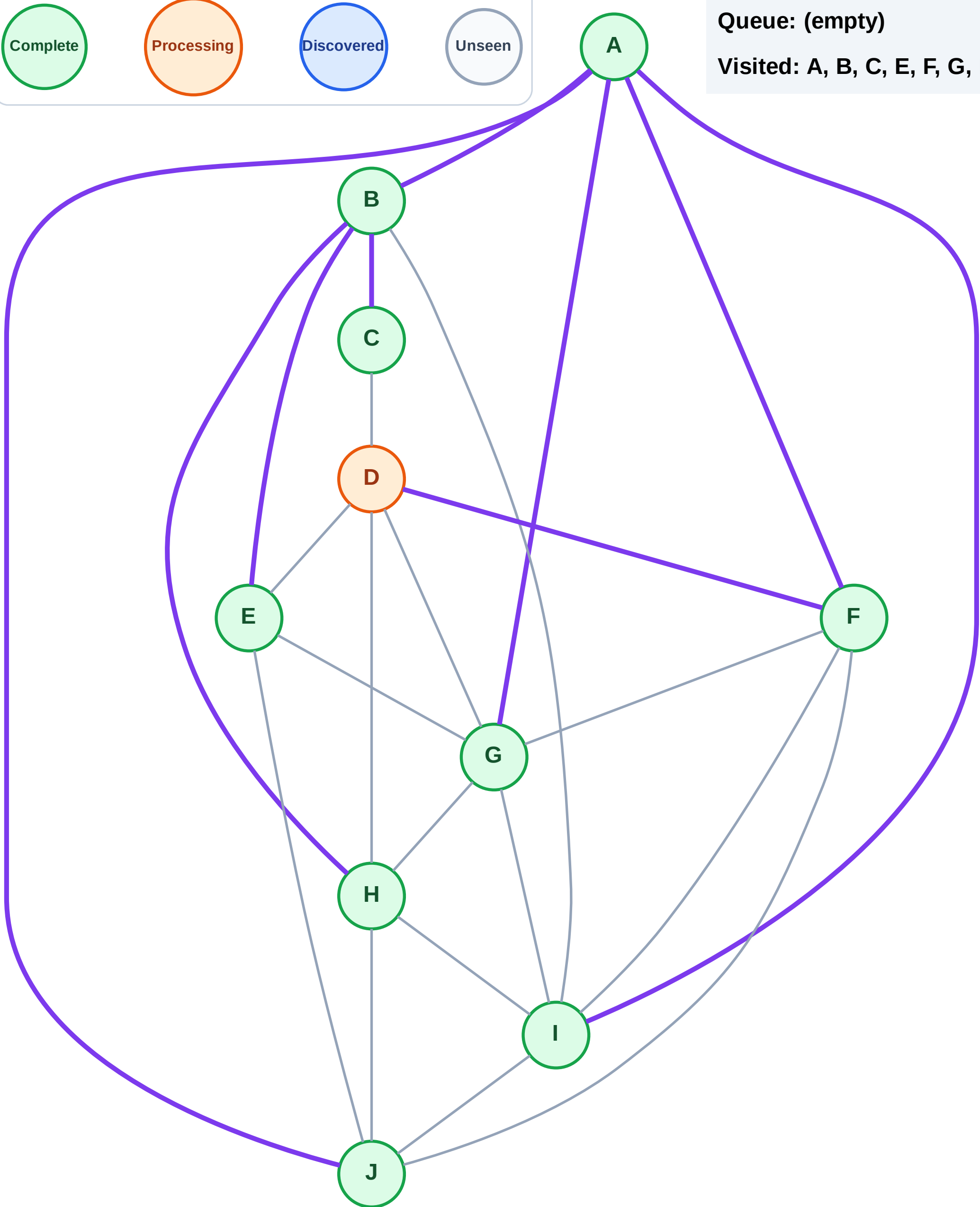
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, E, F, G, H, I, J

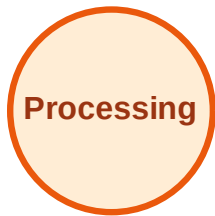


Step 21: No new neighbors from D

Step 21: No new neighbors from D

Complete

Complete



Processing

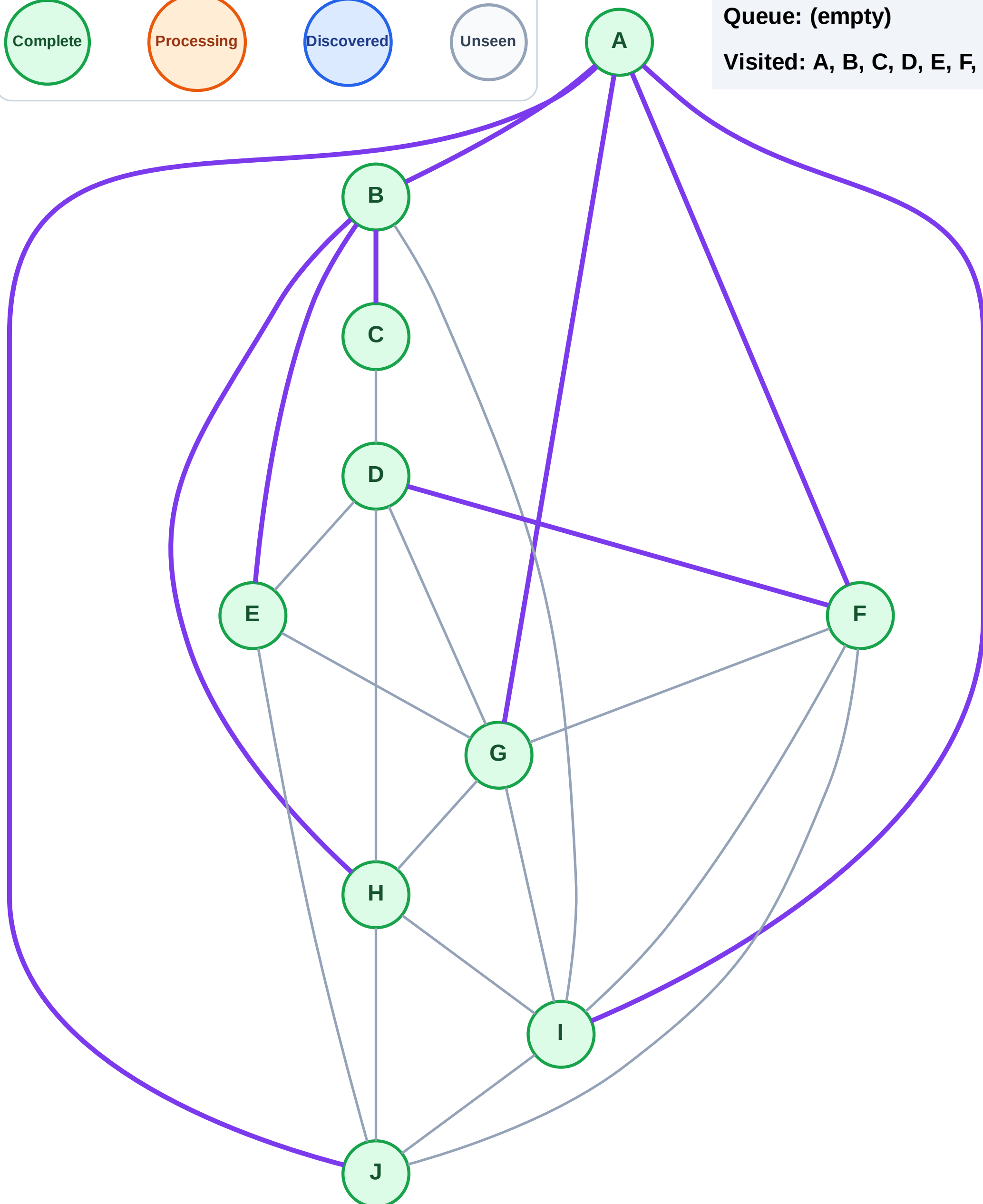


Discovered



Unseen

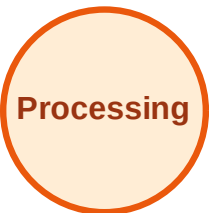
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

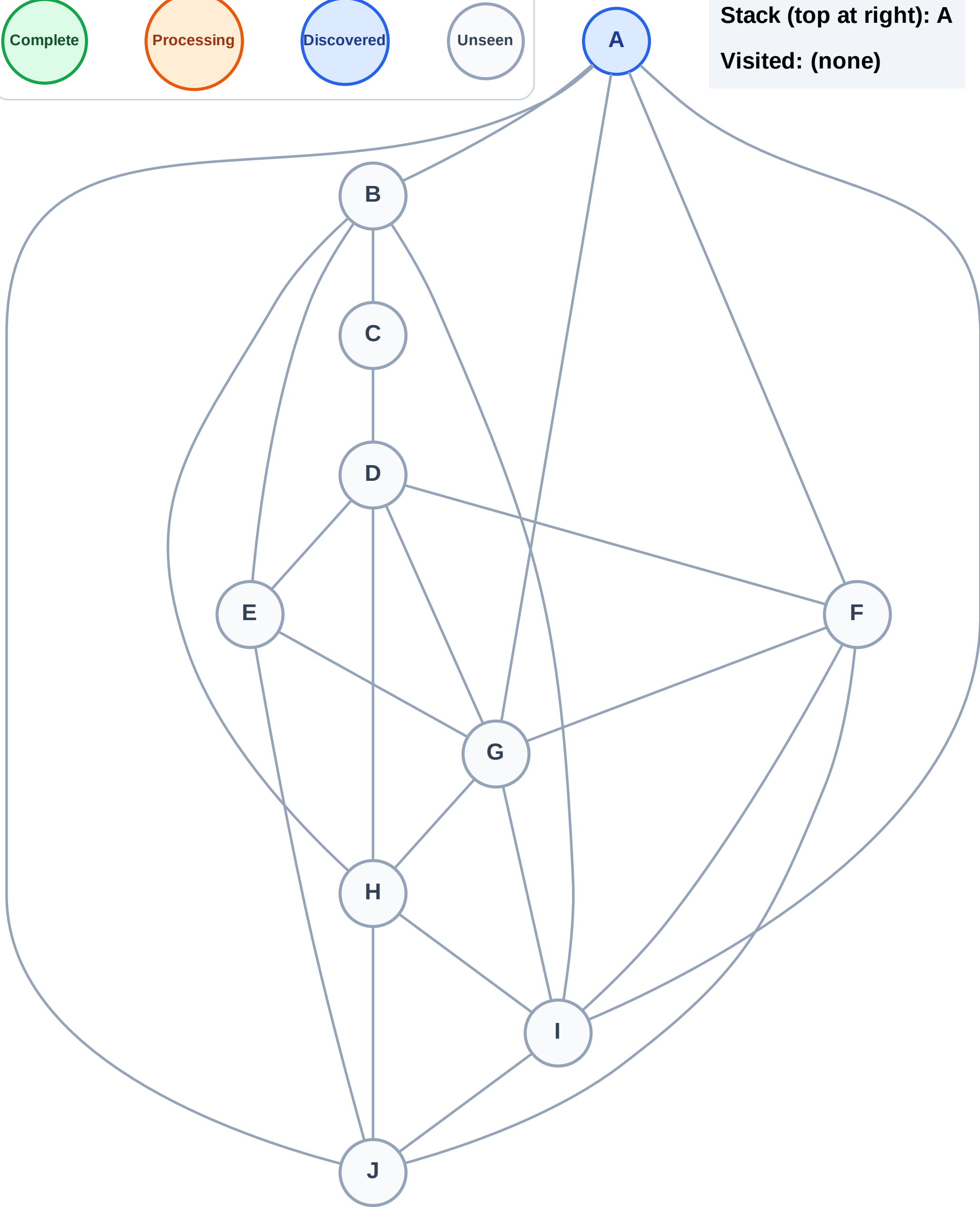
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

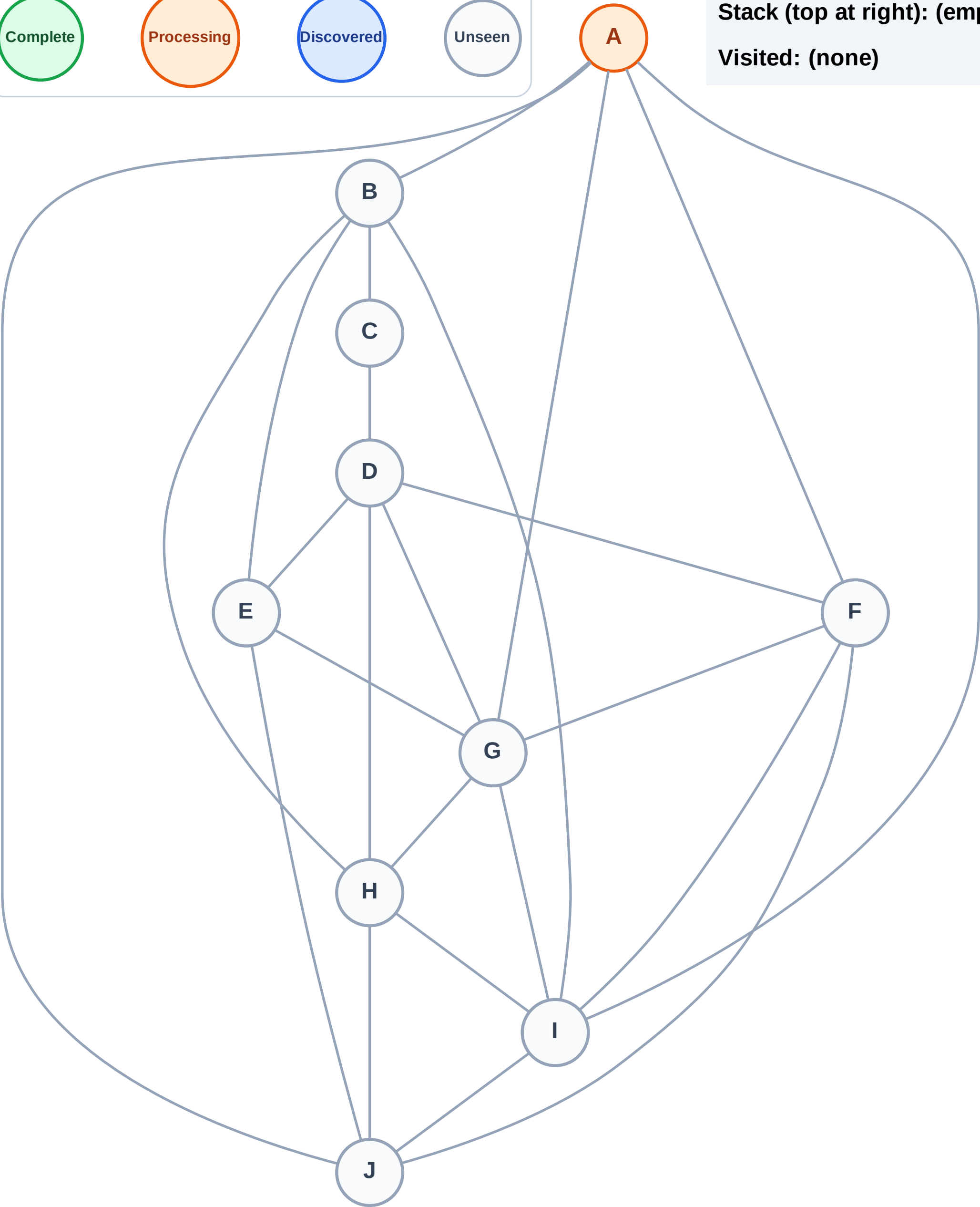
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, I, G, F, B from A

Legend

Complete

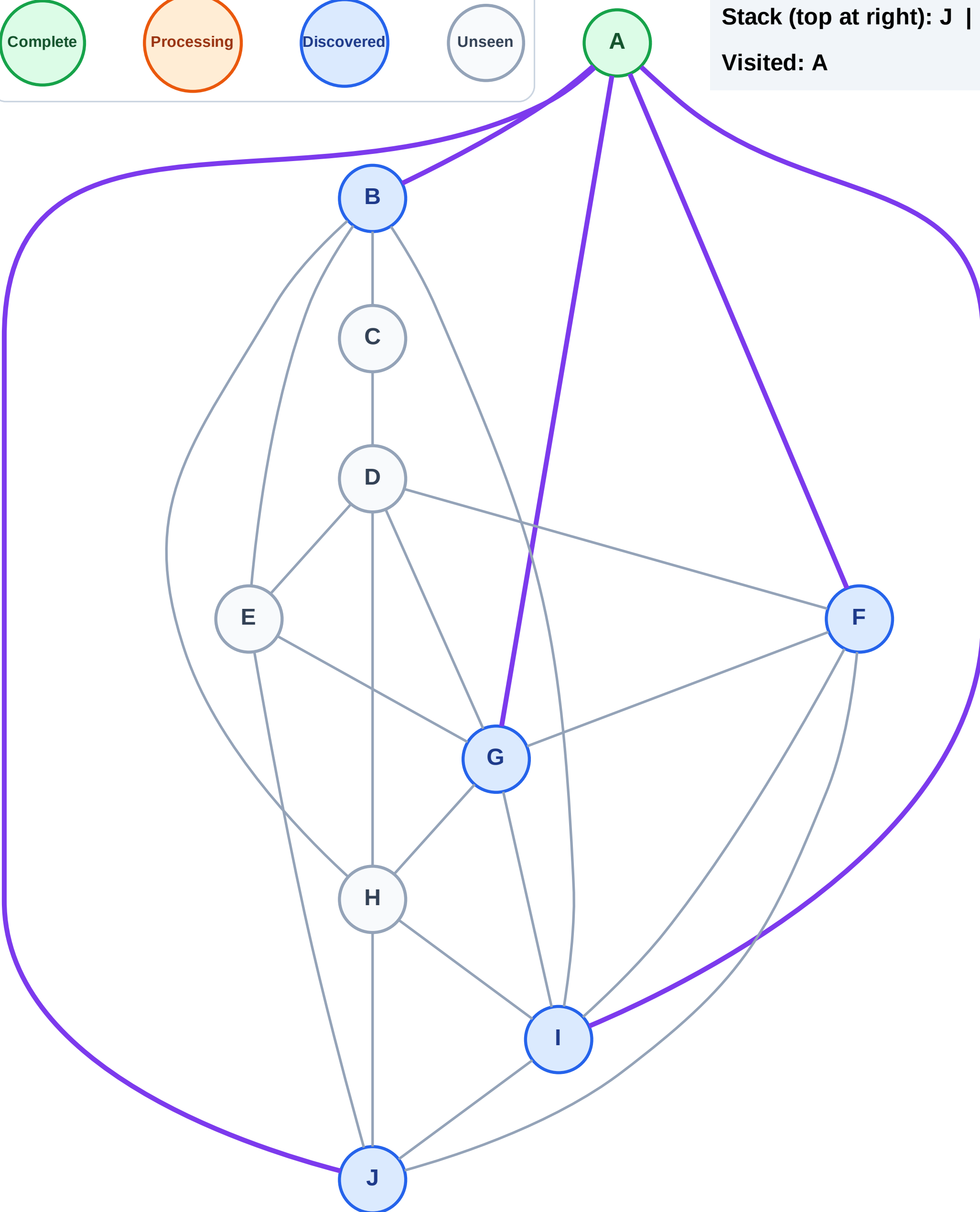
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

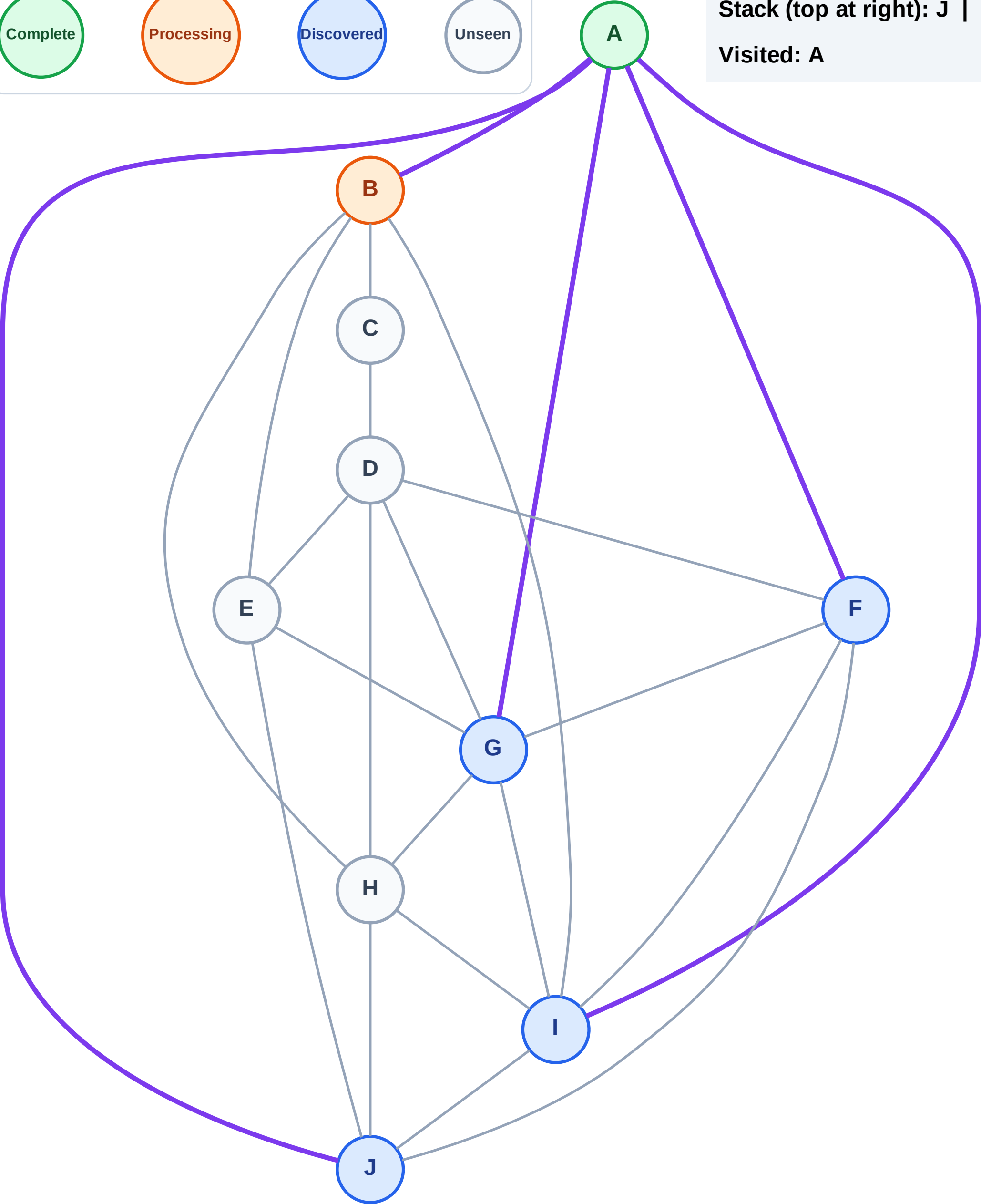
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F

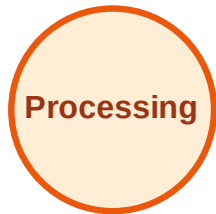
Visited: A



DFS Traversal

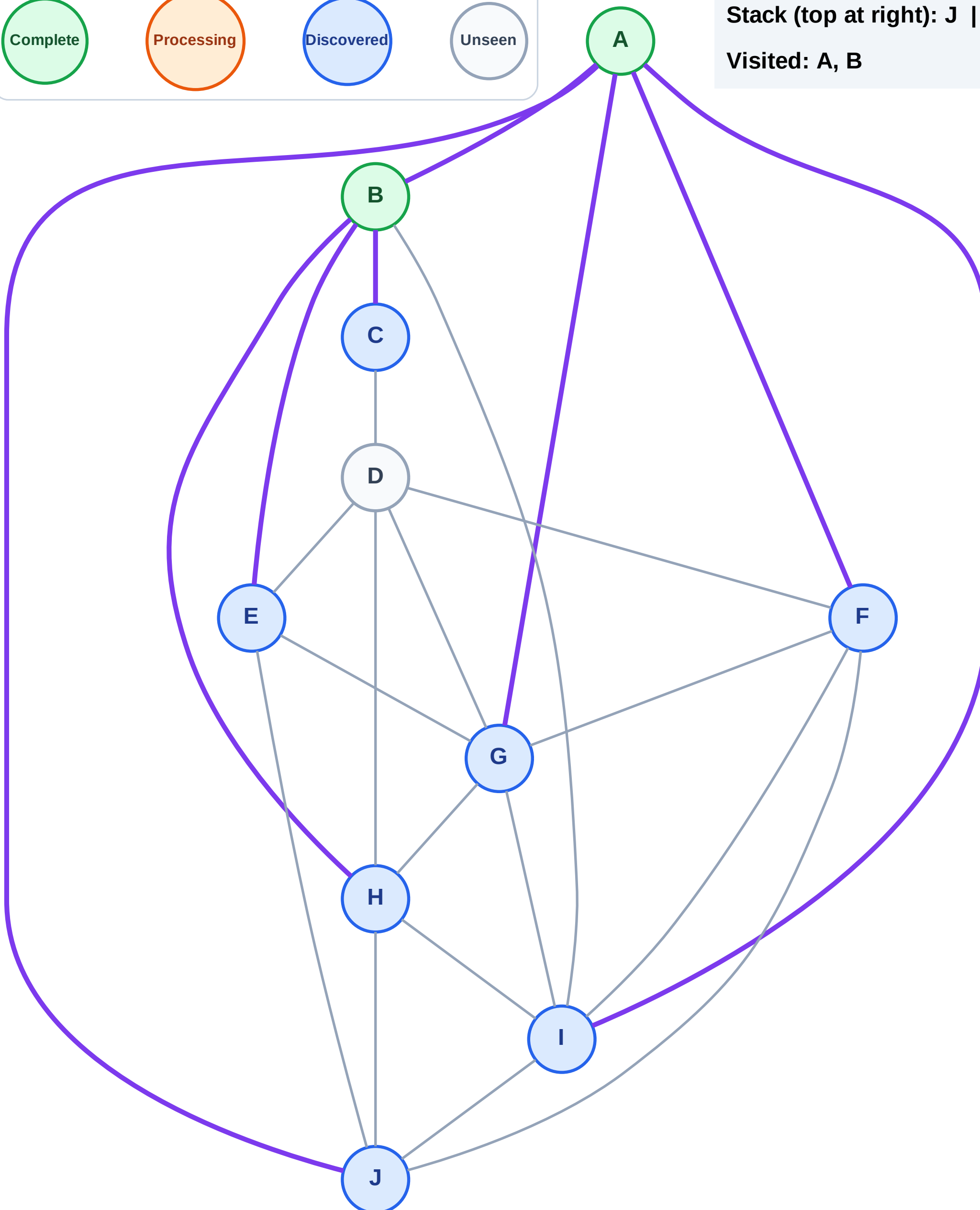
Step 5: Discover H, E, C from B

Legend



Stack (top at right): J | I | G | F | H | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

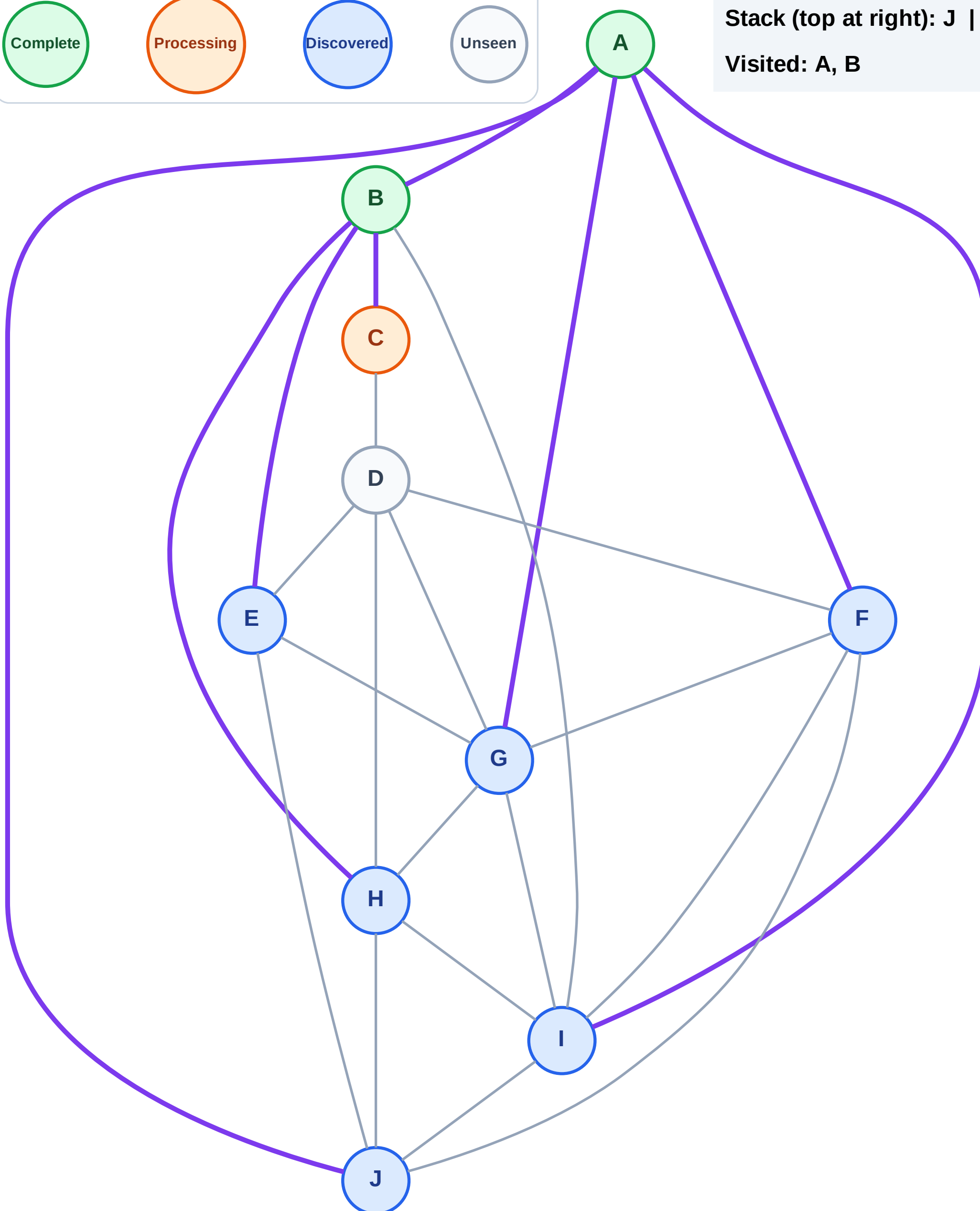
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | H | E

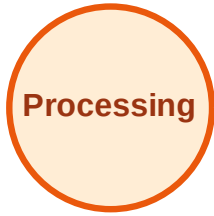
Visited: A, B



DFS Traversal

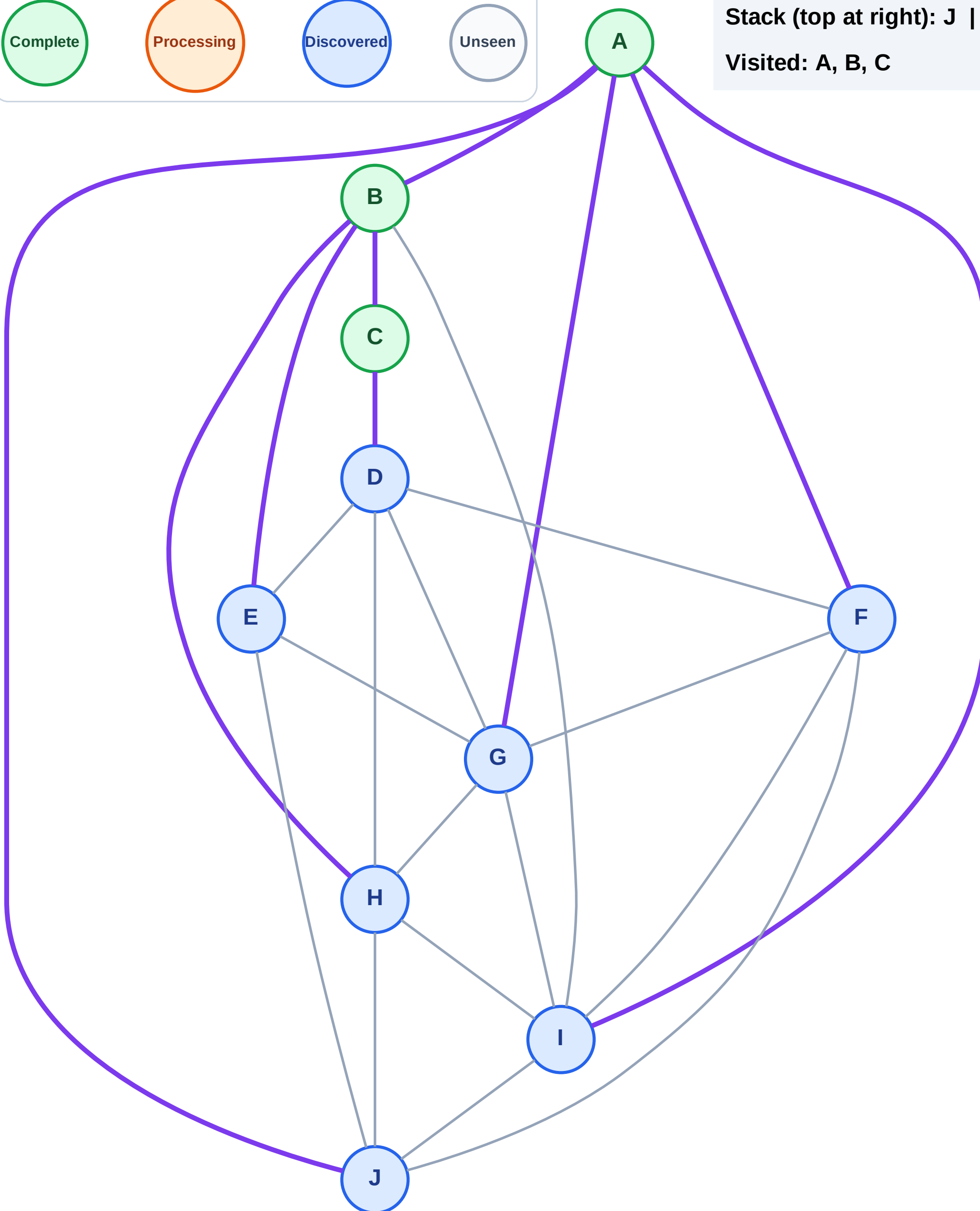
Step 7: Discover D from C

Legend



Stack (top at right): J | I | G | F | H | E | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

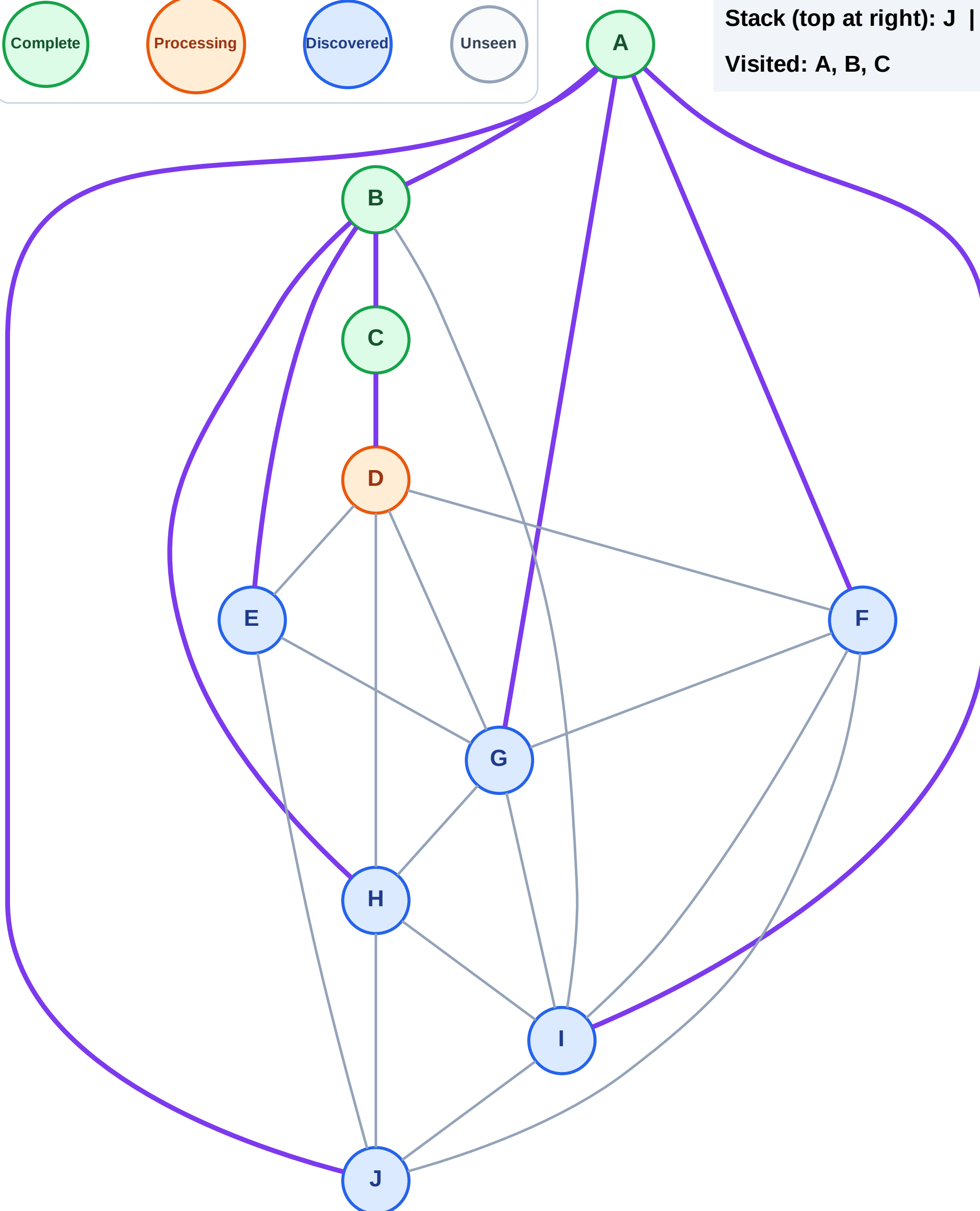
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | H | E

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from D

Legend

Complete

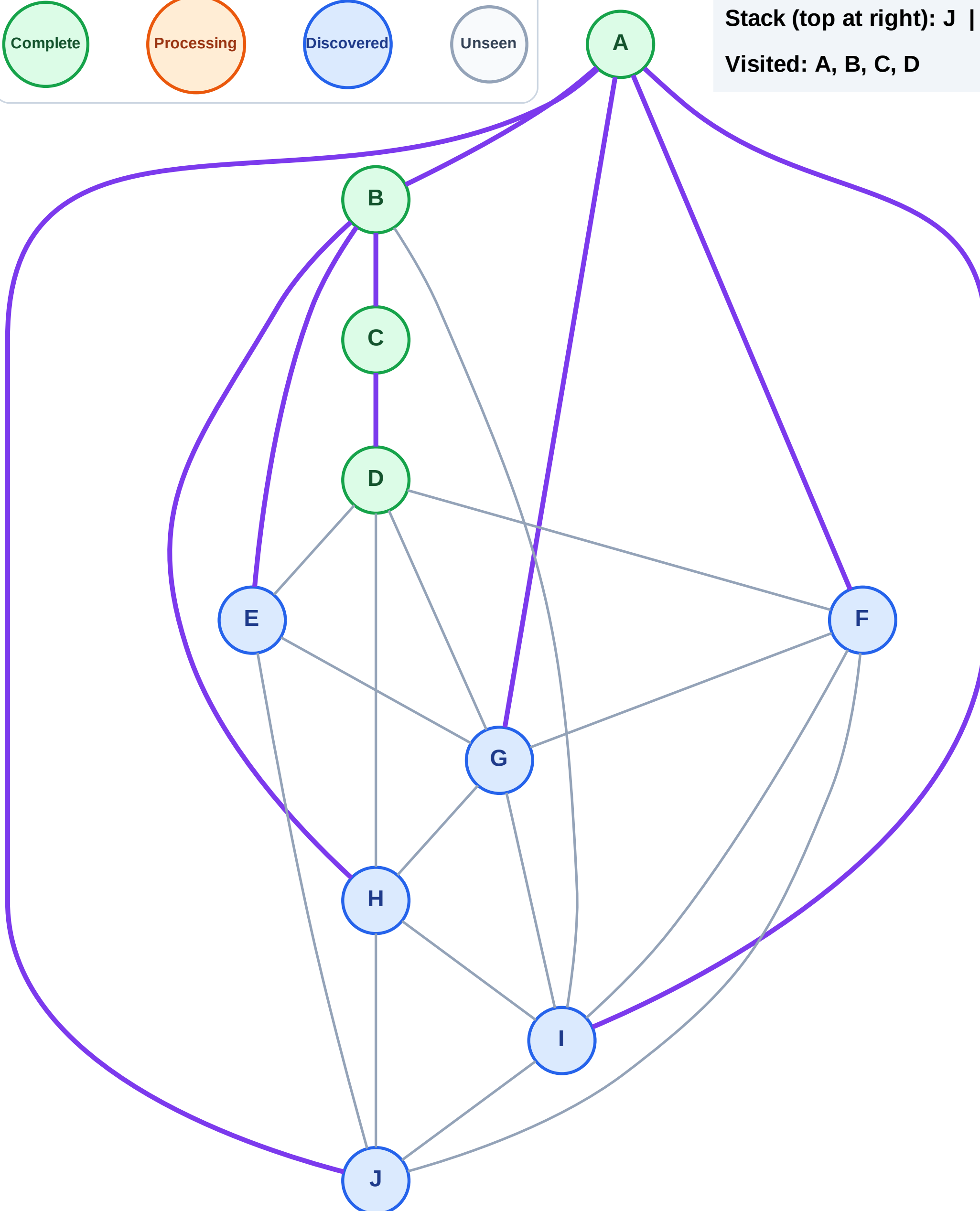
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | H | E

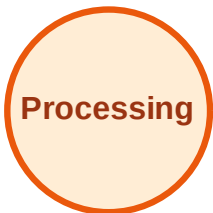
Visited: A, B, C, D



DFS Traversal

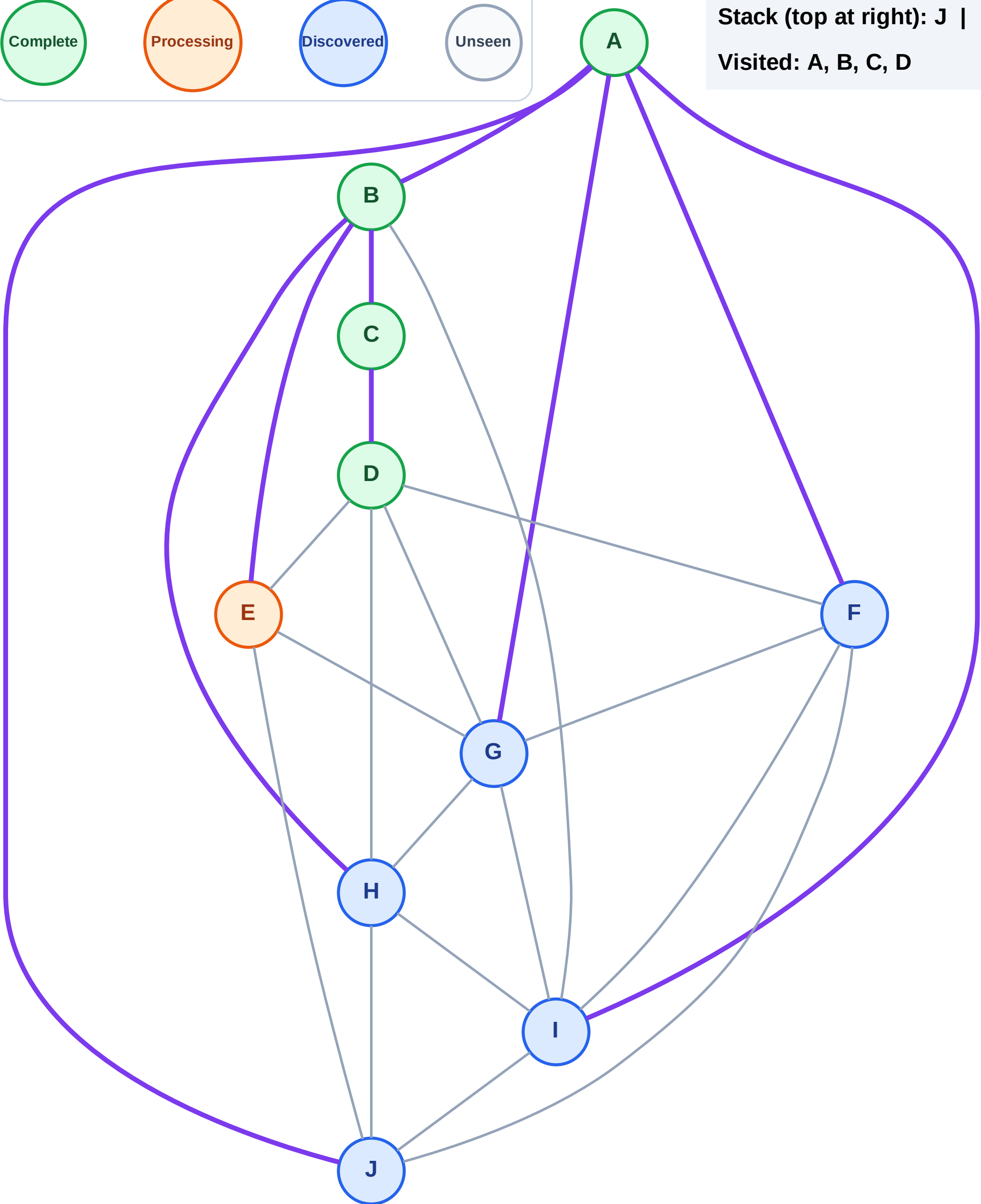
Step 10: Process node E

Legend



Stack (top at right): J | I | G | F | H

Visited: A, B, C, D



DFS Traversal

Step 11: No new neighbors from E

Legend

Complete

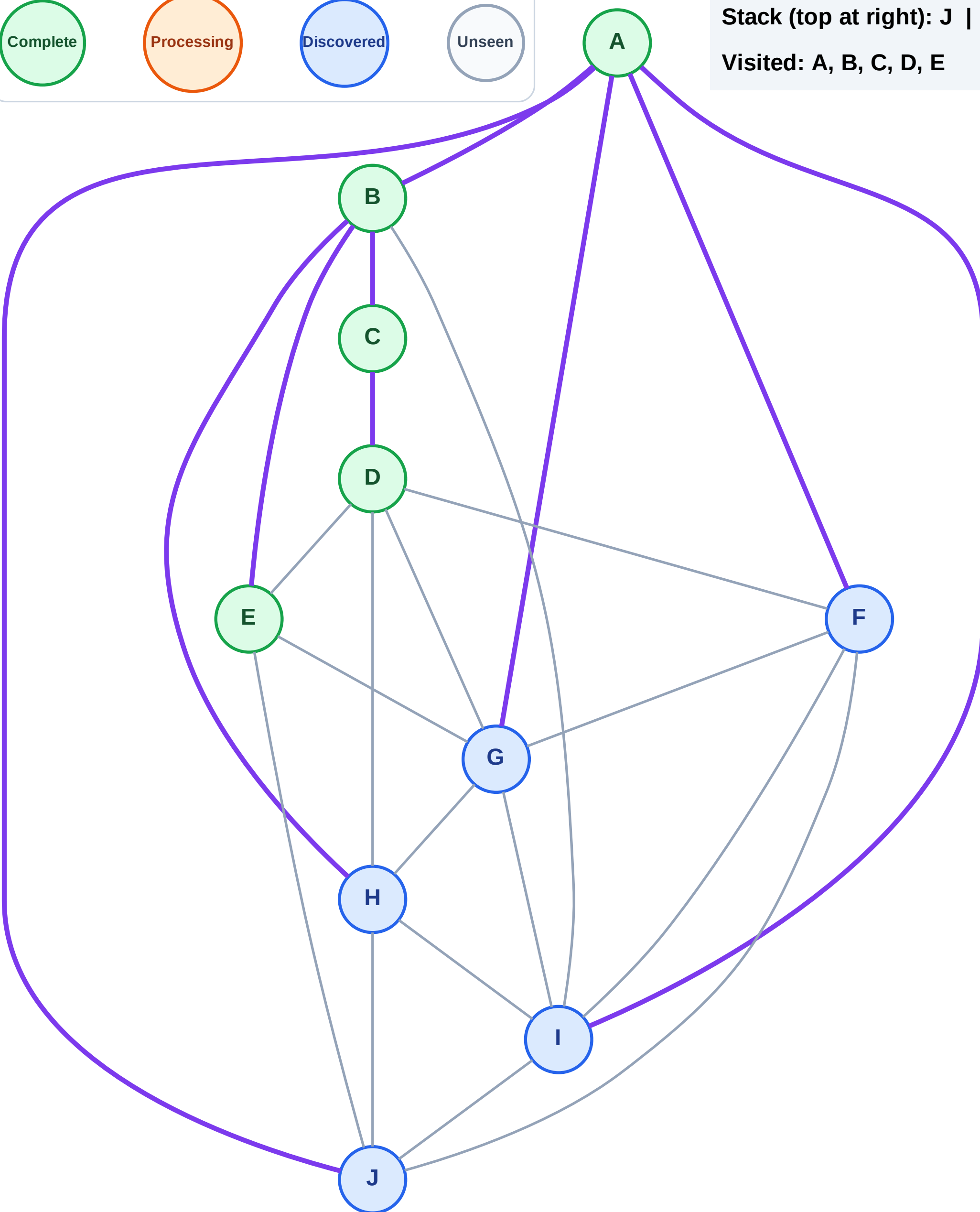
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | H

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node H

Legend

Complete

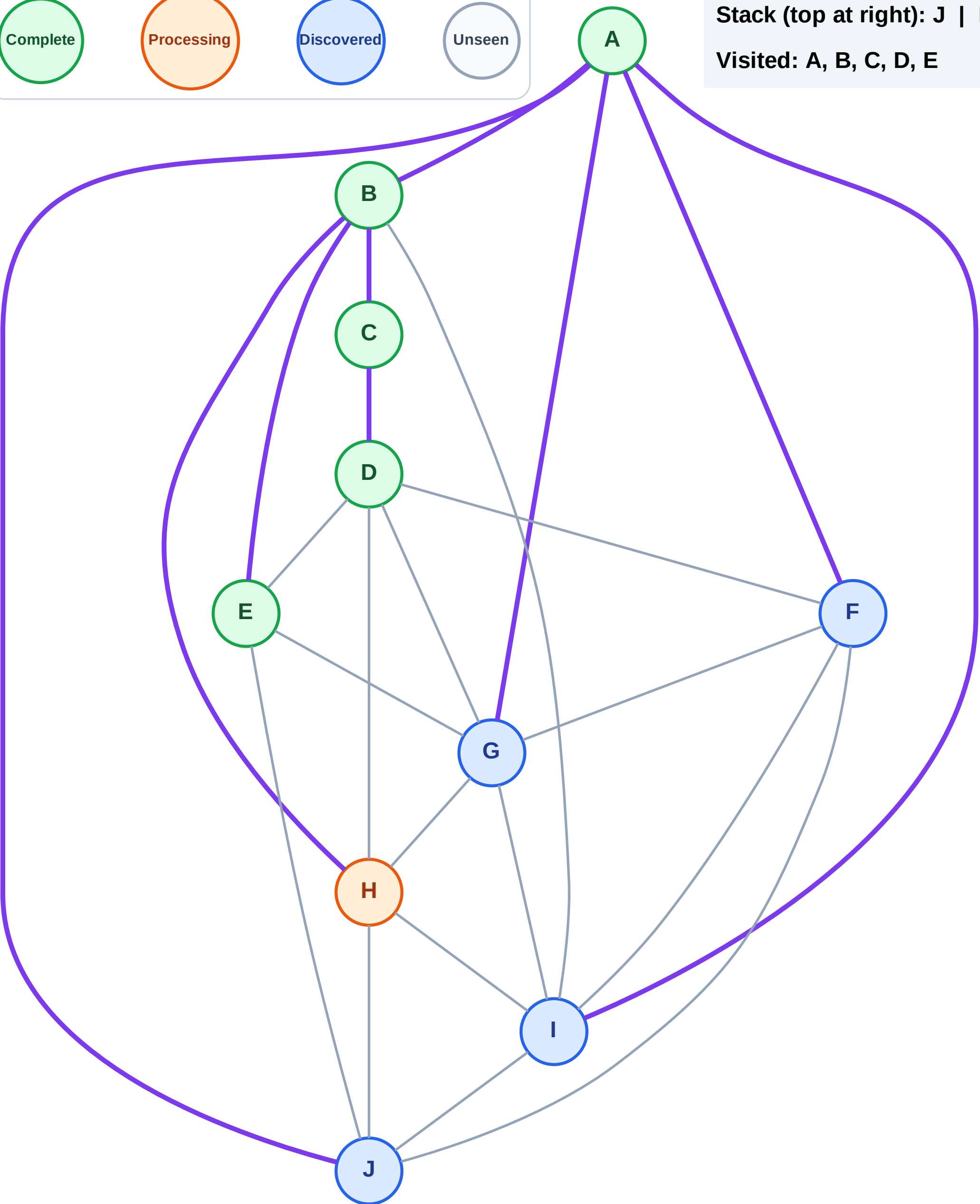
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F

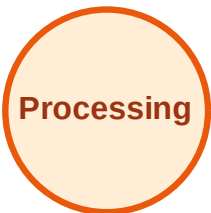
Visited: A, B, C, D, E



DFS Traversal

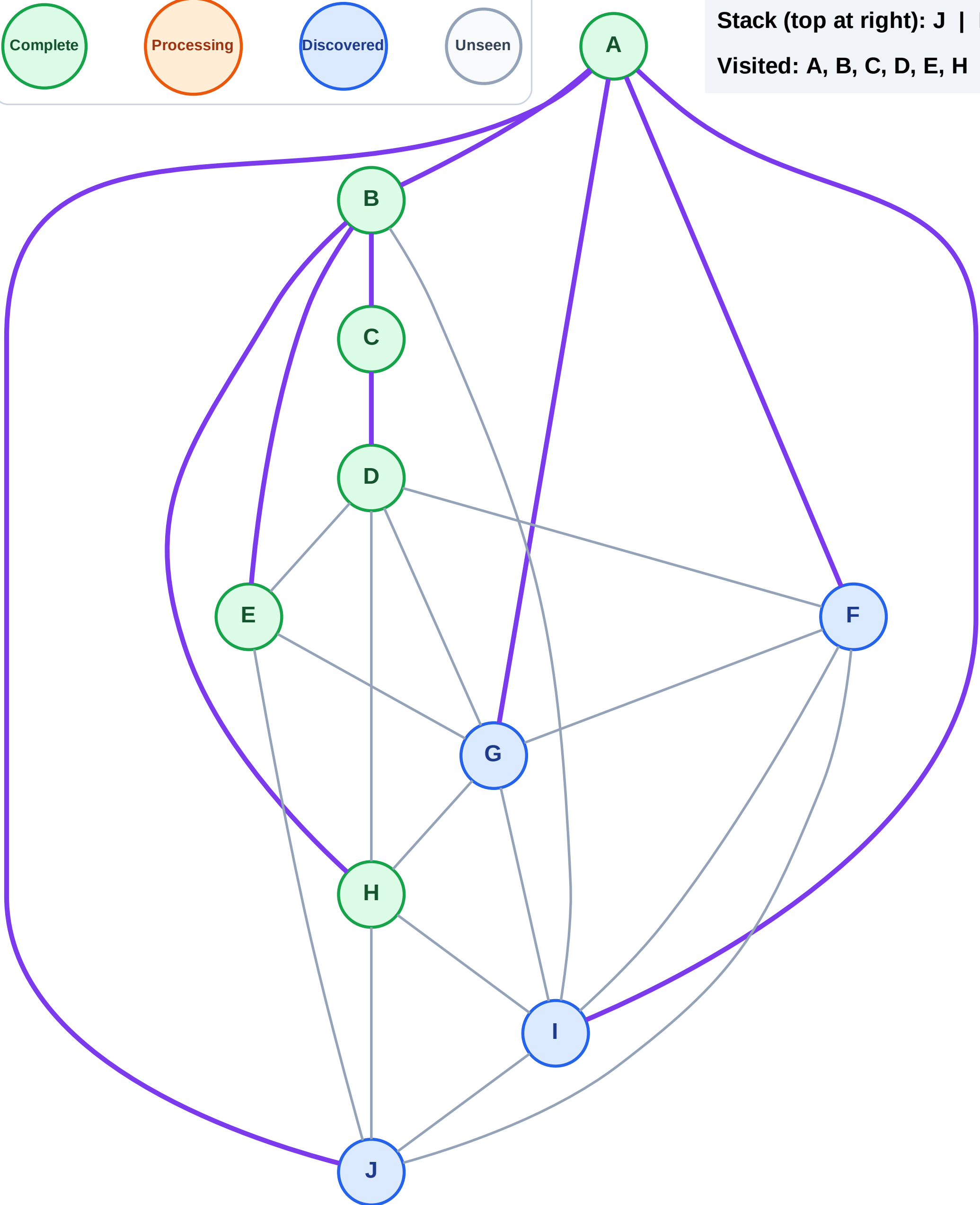
Step 13: No new neighbors from H

Legend



Stack (top at right): J | I | G | F

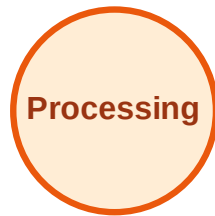
Visited: A, B, C, D, E, H



DFS Traversal

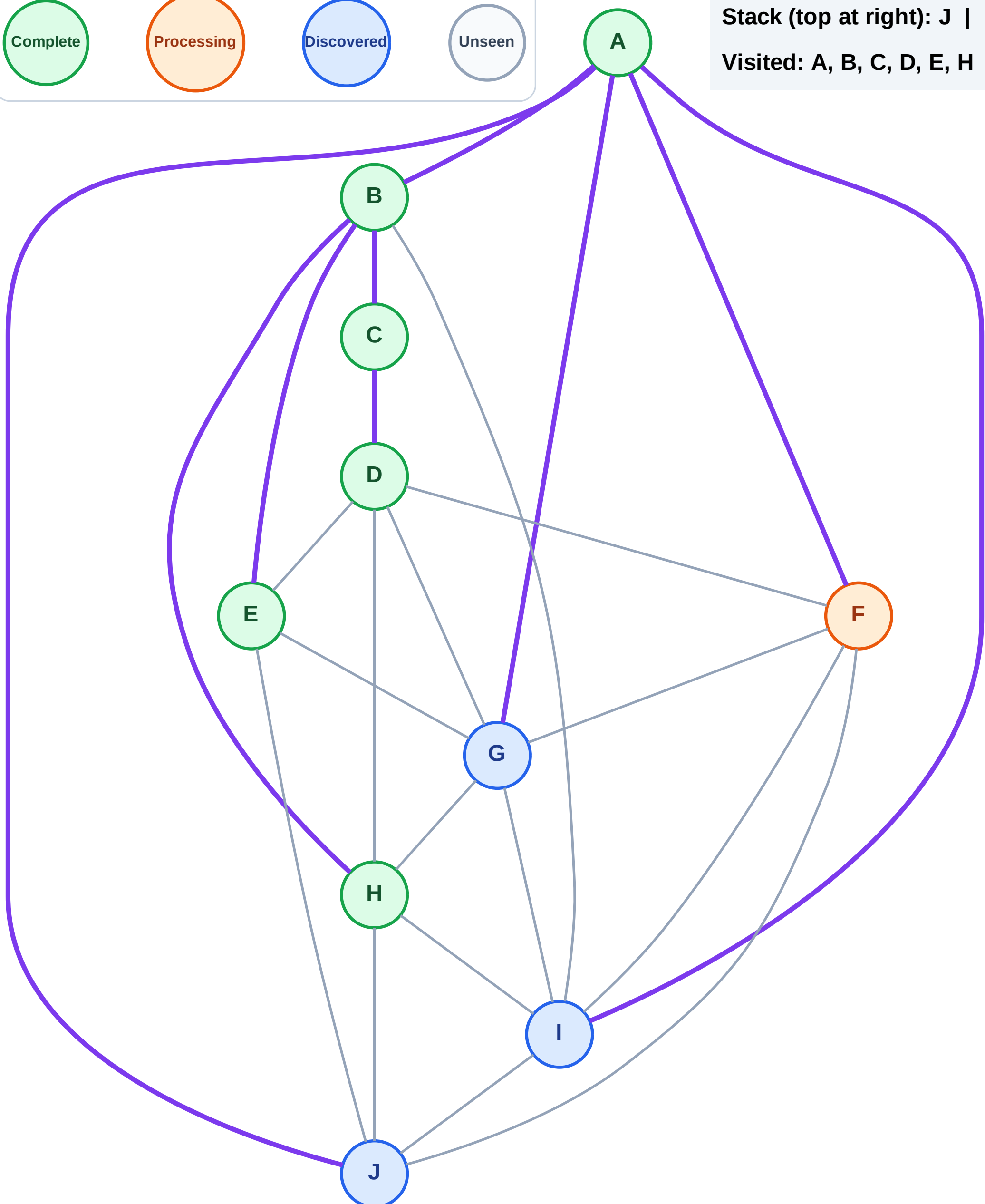
Step 14: Process node F

Legend



Stack (top at right): J | I | G

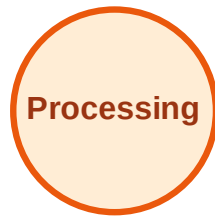
Visited: A, B, C, D, E, H



DFS Traversal

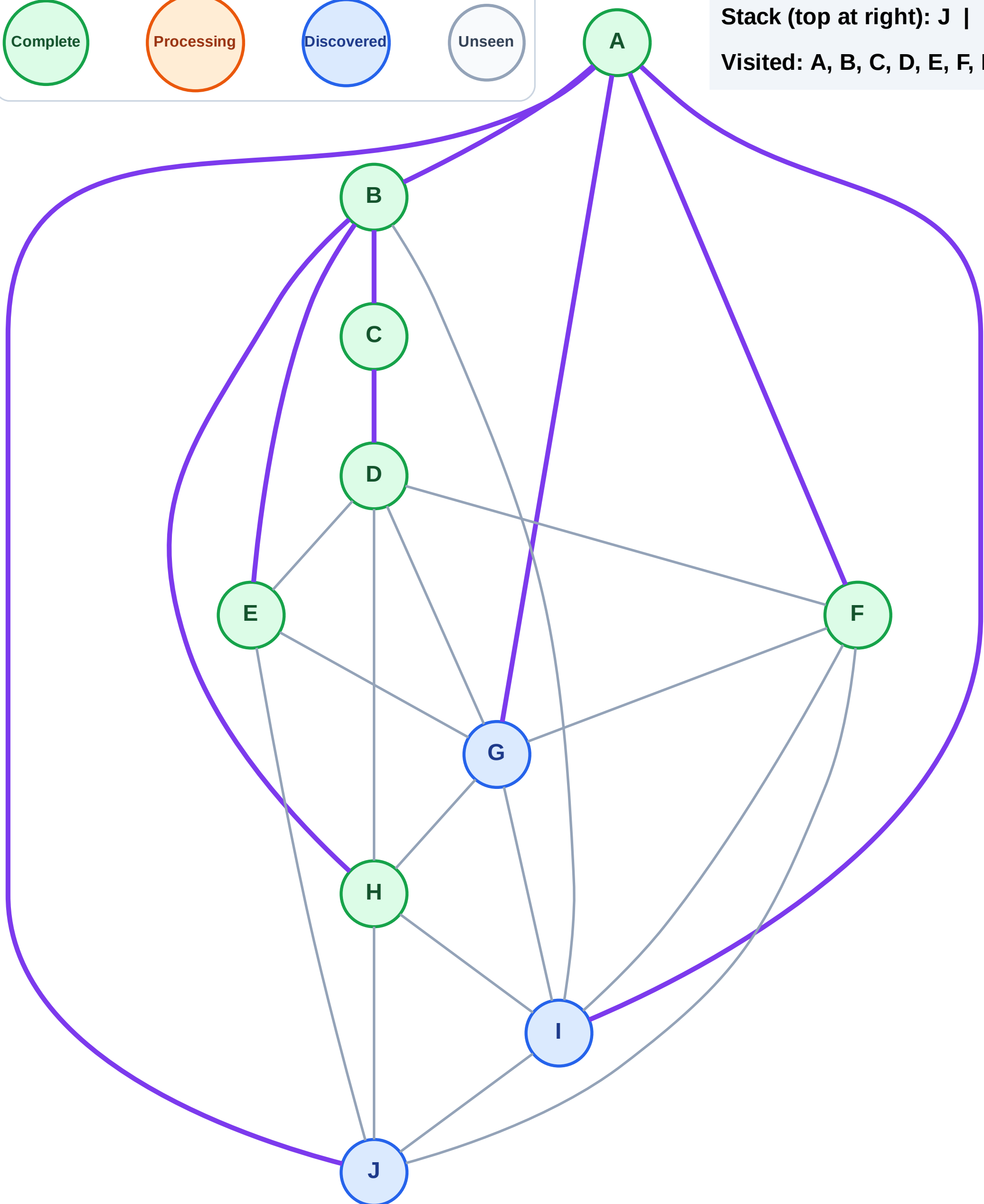
Step 15: No new neighbors from F

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node G

Legend

Complete

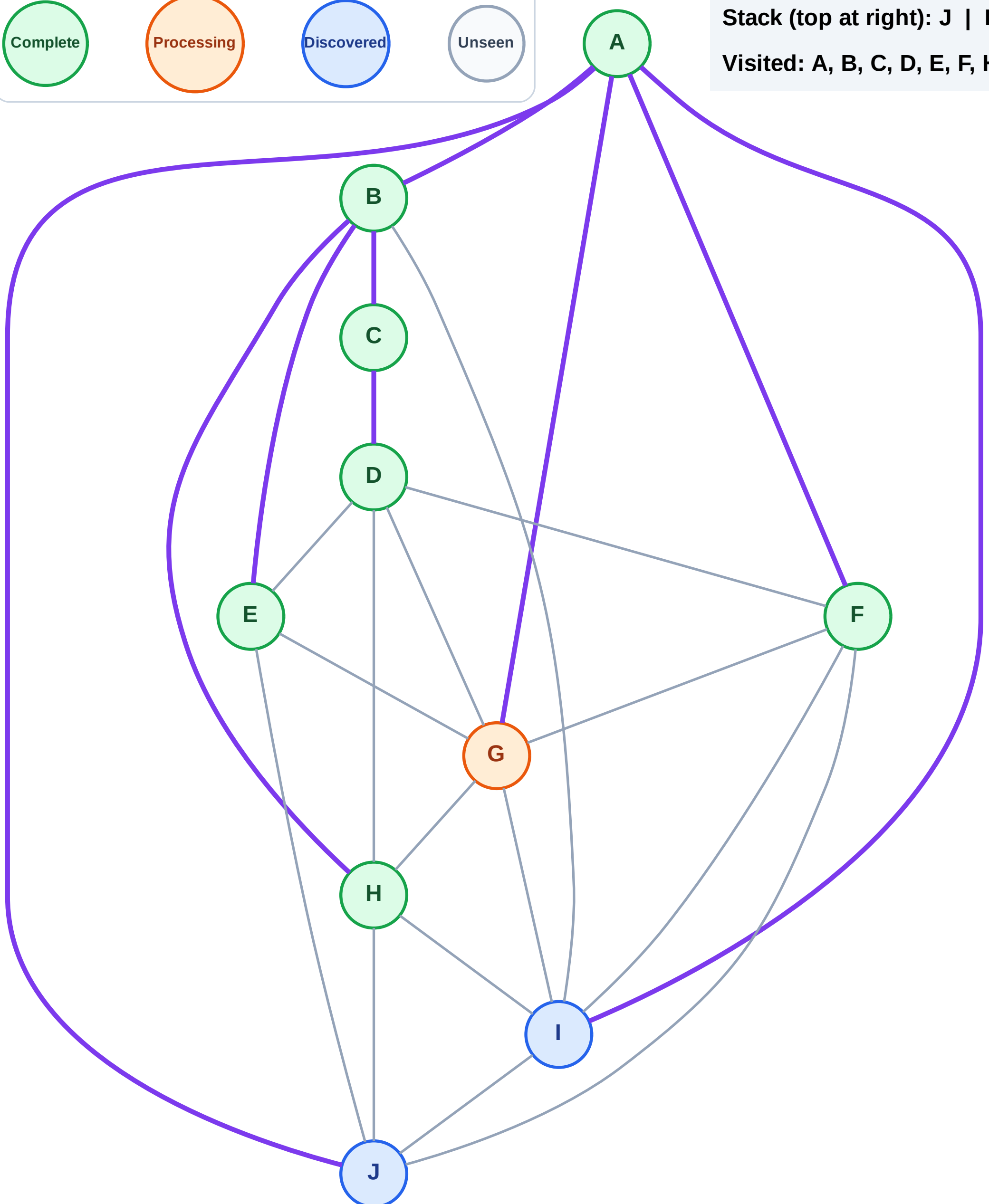
Processing

Discovered

Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, H



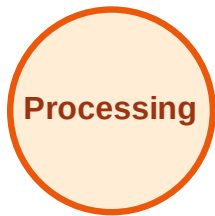
DFS Traversal

Step 17: No new neighbors from G

Legend



Complete



Processing



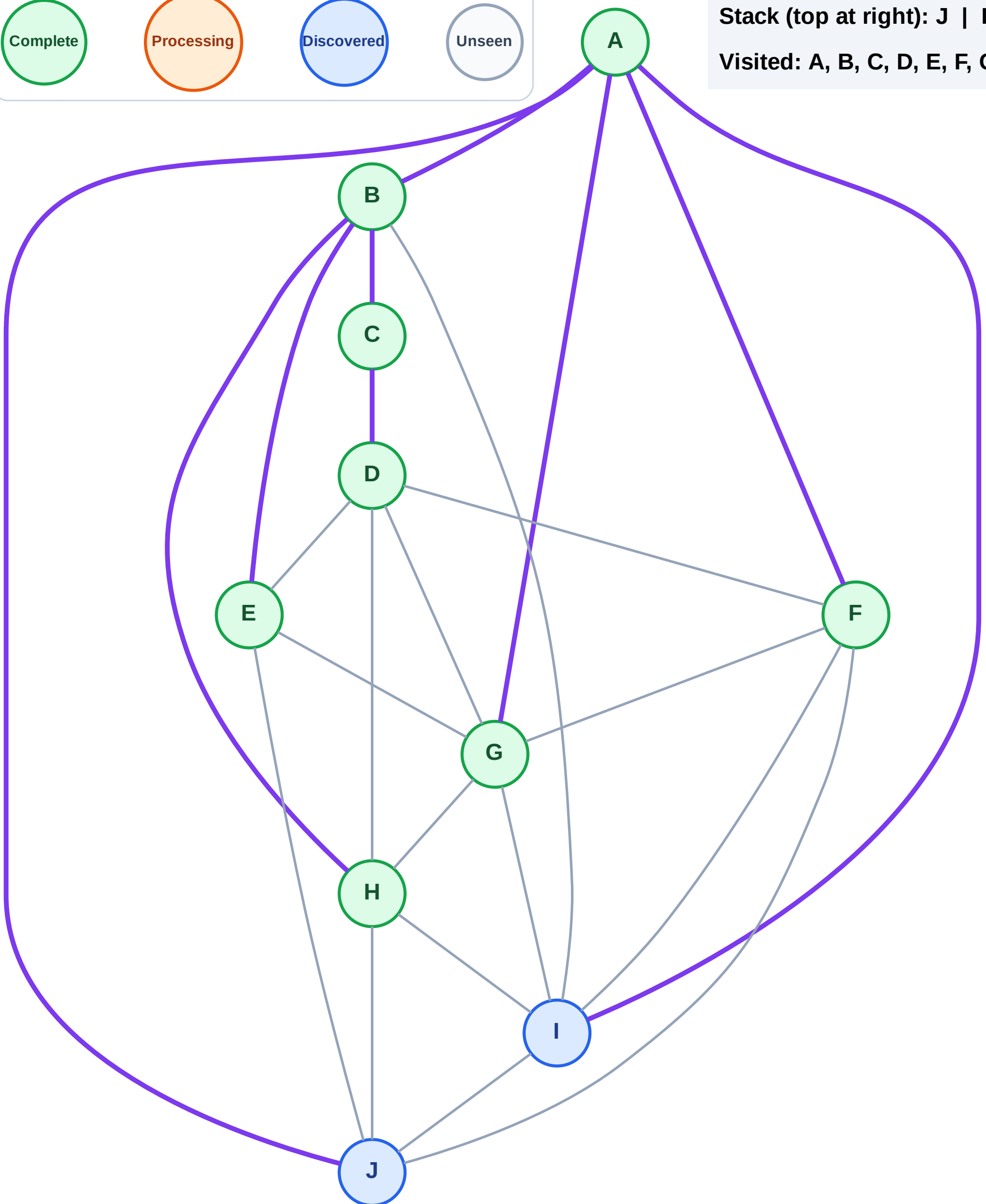
Discovered



Unseen

Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



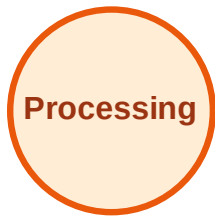
DFS Traversal

Step 18: Process node I

Legend



Complete



Processing



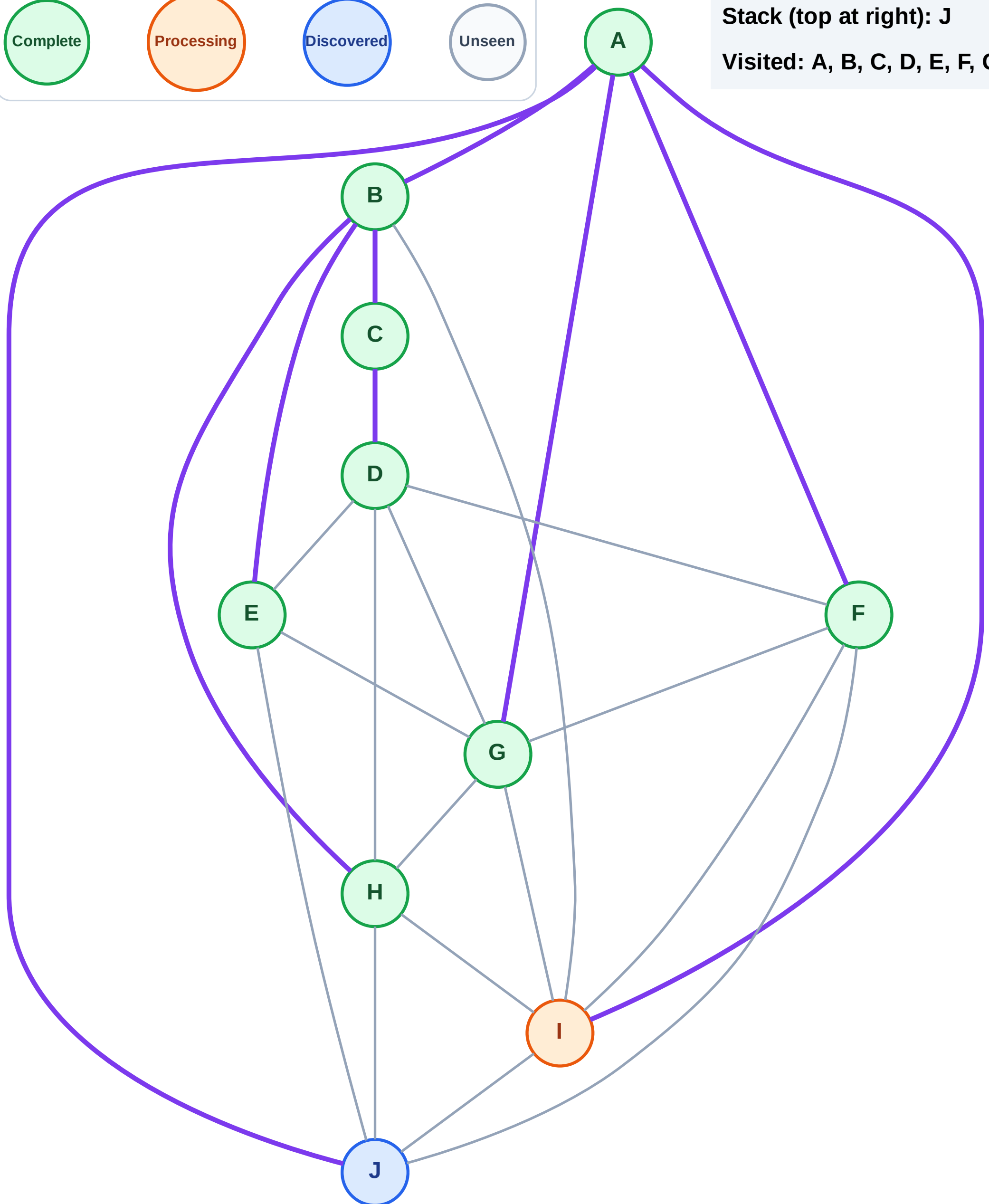
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



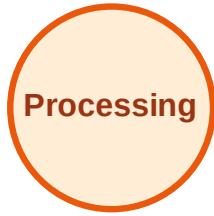
DFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



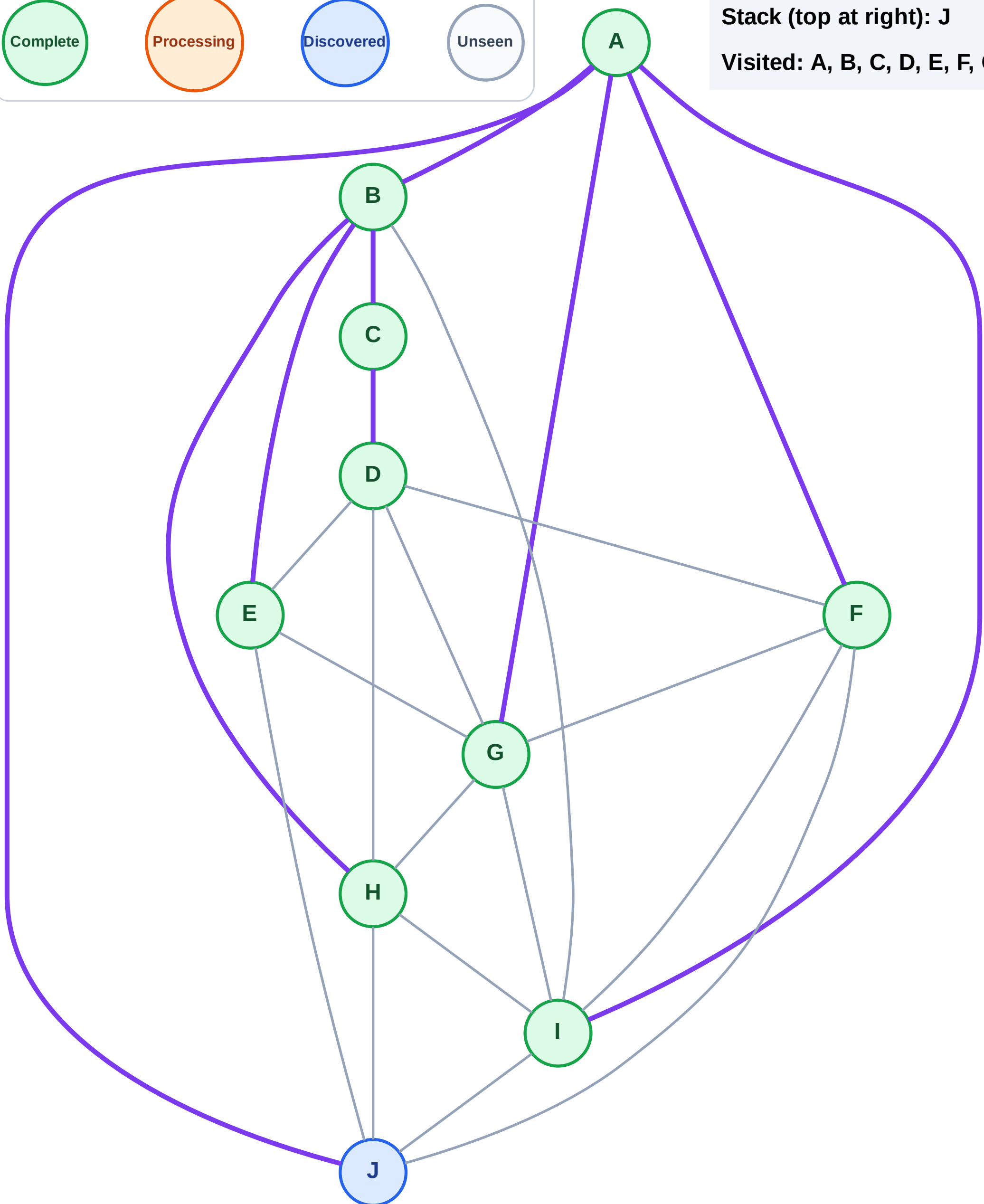
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



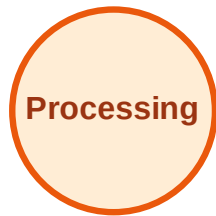
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



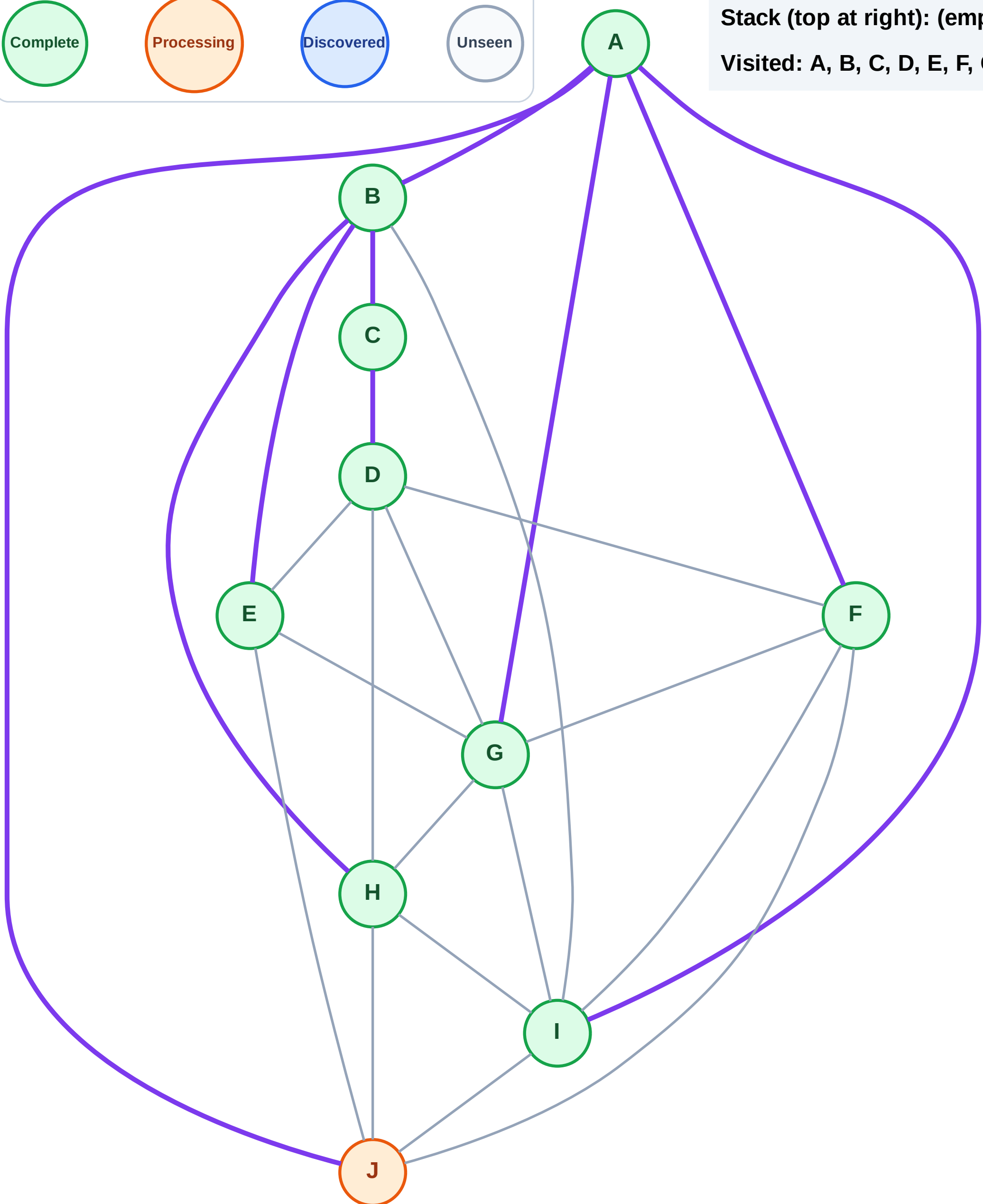
Discovered



Unseen

Stack (top at right): (empty)

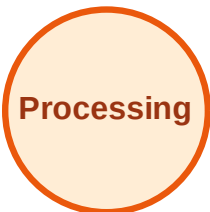
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

